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raw material prices; supplies hit



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# REFRACTED GROWTH

The refractory industry in India, valued at ₹7,000-odd crore, is perhaps not a huge one but the irony is that even a slight supply tremor felt here can bring about a tectonic impact on the steel industry which consumes 70-75 percent of the material, as is the current scenario proving to be.

First, and importantly, clampdowns in China are having far-reaching impact, almost threatening to throttle refractory supply to India and recent developments in dragon country are a pointer towards an uncertain future for the next year or so for refractory makers, which translates into an ambiguous outlook for the steel industry, in a cascading effect.

Secondly, the sluggish state of affairs in the iron and steel industry has definitely impacted the refractory market. This situation is further forcing down the level of capacity utilisation of India's steel industry, leaving a gap for the influx of imported material, that makes China a powerful player primarily from China which at present accounts for around 30 percent of India's total steel imports. "As steel accounts for 70 percent of the total refractory consumption, there has been significant impact on the refractory industry too," Hakimuddin Ali, Chairman Imerys India and Managing Director, Calderys, BU India & Middle East, told Steel Insights.

China can call the shots because it has rich deposits of 2 very important refractory raw materials and uncertainty in raw material supplies is one of the major issues at present. As far as challenges for the refractory industry is concerned.

mayor, on seeing pollution reaching alarming levels, today are willing to pay any price to get the refractory.

Consequently, magnesite raw material prices have spiralled, even doubling in some categories. Australia, Brazil and other places where magnesite is available, were not prepared for this crisis. They had restricted themselves to small production volumes since competing with China is a tall order. Therefore, it is not possible for them to scale up their mining operations overnight.

May be things will be on an even keel after 6 months or one year till other suppliers come into the market and bring some semblance of stability. But the prices of magnesia carbon remain high.

"If steel plants don't get magnesia carbon bricks, their steel plants will shut down. It will take a heavy toll on the Indian economy.

It hasn't happened yet but running the plants on whatever inventory is there in the pipeline won't work," Anirbandip Dasgupta, Senior Executive Officer, the Indian Refractory Makers Association, told Steel Insights. Bauxite calcination in China was also reportedly done through very crude methods which is difficult due to lack of research and development on the same in India. If raw material sourcing is hit, not only steel, but aluminium, copper and cement pollution. Thus, the government has banned industries will also suffer, remarked an

He adds: "Refractory makers are trying to get magnesite from other resources like Australia, and Turkey. But what happened in China was unprecedented, taking people by surprise. But, we in the industry had been warning that such unprecedented events can happen and it is always difficult for the global community to predict China's policy decisions. But things are pretty serious and this volatility will continue for at least a year," he observed. So, refractory makers expect the current scenario to lead to a difficult period because there will be speculation and a consequent spike in prices. The larger steel majors will not be so badly hit because they can still afford to pay a lot more to get the material but it will be difficult for the medium-scale companies.

Refractory makers impacted Impact on prices  
And, the rising price trends of these raw materials, mainly on account of China's stricter environmental policies, have also led to closure of many Chinese manufacturing units of refractory, leading to a further scarcity in the domestic market. There had been a glut of refractory coming to India. Prices of both magnesia carbon and bauxite had gone down in China about a year back by 15-30 percent because of a lack of demand. Supply of magnesia carbon will remain an issue because 42 mines have amalgamated but under government pressure. They have not united their operations as per their free will. So, co-operations amongst the mine owners is not expected. "Each player would want the other person's mine to be opened first so that he can conserve his own reserves. There is bound to be conflict and, in such a scenario, supply will be erratic. And the amount of magnesite mining that has been stipulated per annum by the government is not enough to meet global demand for magnesia carbon," Dasgupta adds. In the domestic market. There had been a consequent inventory pile-up in China that goaded miners to sell at much lower prices. But now, in a sharp reversal in trend, there is a 40-50 percent rise in prices. Today, magnesia carbon prices, under normal circumstances, which are ₹60,000 per ton, are touching an astronomical 1,00,000-200,000 lakh per ton! And refractory makers say they are passing on the cost to the end-users (read steel industry) because their hands are tied. On average, refractory prices have increased by 60-70 percent from May till October. But it depends on the product and policy. They now need to work around the raw material. For instance, some users need silica, which doesn't require any imported raw material since it is readily available in India. The gap remains and, maybe, we will be able to get a clearer picture of how much crude steel production has been hit, in the next 6 months," said an industry source.

#### Bauxite

The second factor impacting Indian refractory makers is bauxite, China also enjoys rich deposits of refractory-grade bauxite. The bauxite deposits available in India are metallurgical grade whereas the category available in China has a higher level of alumina, of around 80 percent. Hence, refractory makers in India have to import most of this material. Whatever is available within India contains a lot of impurities and is thus economically not viable.

"We have limited supplier base for refractory grade bauxite and, in the absence of any new mining and exploration activity, its availability has become even more uncertain. Most of the bauxite currently being mined is not of refractory grade and goes to the metal industry. Refractory grade bauxite in India is limited to parts of Madhya Pradesh and Gujarat and, of late, the quality of refractory grade bauxite is further deteriorating on account of increased Fe percent," says Ali.

India also has good deposits of sillimanite sand.



within India. But where 90 percent of the material required has to be imported, then obviously prices are going up to astronomical levels. In fact, in some cases, prices have trebled. Consequently, finished refractory prices in India have gone up from 60,000 per ton to 2 lakh per ton. And, sources say, these are being quoted to very large customers who have tremendous bargaining power in the market and normally arm-twist suppliers to come around to price levels of their liking. Such suppliers are taking the lead in absorbing the price rise, saying they want the material at any cost. They have no choice since their production will come to a grinding halt.

So far, supply has not been disrupted. And there has not been reports of any loss in steel production due to refractory supply

issues. But there is panic in the market the China crackdown, plans are being made raw material at say 2,000 RNB per ton. Then say industry circles. There is no inventory for 6 months – I have 2 months of inventory they book the order as per the raw material in in steel companies' yards, they add. There is stock, there is some volume in transit their hand and manufacture likewise. They is no commitment on supply either. "The and some have been committed to the user close the contracts in 7 days' time. Thus, refractory end-user has managed and industries. So, nobody was impacted for the working in the current pressurised scenario, refractory suppliers have also managed last 6 months, because the supply was there they have changed the buying cycle by first by started giving alternatives. Because the pipeline. But things are at the edge now buying the raw material and then booking manufacturers have to look for alternatives if that six month's inventory is drying out. the order.

the scenario in China persists to work around Now, China is not clear about its environment In India, however, refractory makers the problem," they say. plans. It is not supplying the dynamite which are placing orders on the basis of average

A senior official at a leading refractory is required for mining. There is no clarity consumption. Players are booking orders maker said: "We are working on good after the Communist Party of China (CPC) which are valid for 3-7 days. And they are alternatives. But, there is panic for steel-making. Winter is approaching. Peak winter pledging to supply within a month. Then makers in terms of one, refractory scarcity will be from December-February. We are again, they are looking to book orders. and two, the high cost." now wondering whether this scenario is the Nobody is accepting long-term contracts.

The official added: "We, in this situation, worst or whether things will worsen further!" The delivery period is 30 days. The order are not being able to make money because the "So, the immediate second half is very must be finalised in 3-7 days. And this pricing is on the spot market. For instance, uncertain. Before April there will be no has improved from 24 hours to 7 days, say steel maker is going in for long-term orders question of stability, as per experts at a recent industry sources!

or annual contracts for supply of refractory global refractory conference held in Chile. Another downside for refractory makers or annual contracts. We are booking orders looking at the Chinese supply scarcity, is that government companies have forged expecting that the raw material costs will European suppliers and elsewhere are taking annual contracts with the refractory makers shoot up by 10 percent. But when we go advantage of the situation. Their material at year-ago (February 2017) prices and to the raw material suppliers, we learn that's costlier," said the official, adding, "Every thus the latter are incurring huge losses in prices have shot up further and we have to month, we are writing to our key customers supplying to the state-run companies. The shell out 25 percent more. But the refractory regarding what is happening with regard to state-run companies are liable to invoke makers cannot go back to their clients with refractory raw materials in China. Even if force majeure and if the refractory makers a revised price sheet! They have to absorb they give us the price we quote, availability is do not supply then these PSUs can invoke that extra cost. It is like a gamble at present not secured. The refractory cost for the steel the risk purchase clause in their contracts. Sometimes it does happen that we have makers is only 2 percent. At best, it will jump These prices were contracted on the basis quoted a 30 percent increase in the price and up to 4 percent because of the China factors of the February 2017 rates and at that time have received a 20 percent increase in our raw But if it is not available, then steel production no one had apprehended the turn of events material cost. So, under such circumstances will get affected," the official adds. that have taken place in China. And these we earn 10 percent!" The Chinese refractory makers too are contracts run for 1 year to one-and-a-half-

An official at another well-known working around their own problems. They years, sources indicate. refractory maker said: "The market is one have stopped exports because they want to "We have also highlighted (to the big gamble and the worst is that because of their domestic market first. They buy the Ministry of Steel) issues with regard to the



one-sided public sector purchase policy it has reduced to 11-12 kg per ton of crude of the overall cost of production for steel-making. Currently, we are in the process of generating steel-making. If Japan is any benchmark, which is a pittance but without data to be shared with ministry with an then its specific consumption is 8-9 kg per refractory no steel producer can manufacture objective to secure the future of the Indian of steel-making. Thus, refractory usage is a single ton of steel. And herein lies the refractory industry," Ali said.

Industry sources say within private actually quite steep. Companies are innovating in terms of players, there is better understanding to For instance, one year ago, the ladle replacing magnesia carbon in a bid to survive. preserve continuity in supply through performance was around 150 hits – which For instance, in steel ladle metal zone, delivery assurance, not contracts. means, after one refractory lining, it can be refractory players habitually used magnesia

#### Lower capacity utilisation

After steel, the other main industries that hits today. Refractory longevity means less quotient. As already discussed, this material consume refractory are cement (7-8 percent) usage of the material. is available in other parts of the world too, aluminum and copper (7-8 percent), glass How is the refractory industry, in that like Australia and Brazil, but in lower (4-5 percent), followed by others. case, combating the decreased use of its volumes, and supply from these geographies has not opened up widely due to the cost

The installed capacity in India is around material? "The operational efficiency is increasing structure etc. Magnesita carbon is costlier in 16 lakh tons. But overall capacity utilisation is around 75 percent, as per some sources, but to achieve that players are bringing in the European Union and Brazil compared to while others say this figure is much lower and new kinds of refractory which is costly. After that available in China. But a few companies, around 60 percent at present because of the all, steel makers cannot get 230 hits by using which have good R&D, are bringing in spinel sluggishness being experienced by the Indian the same lining material they used earlier," which is a separate refractory material and is a substitute for magnesia carbon. It is being sourced from Europe and is made through

But, sources at TRL Krosaki say Refractory players are improving their sourced from Europe and is made through that there are some reasons for the low materials through the use of additives the use of a combination of raw materials like capacity utilisation. There is a concept of that may even be synthetic in nature. It is spinel and alumina.

specific consumption, they say, whereby becoming costly, but they say the overall "This refractory is costly but more the performance of the refractory itself is cost for the user-industry will be reduced. cost-efficient. For instance, JSW is getting improving due to increased operational For example, the refractory cost `550 per 230 hits against 130 hits 3 years back. But efficiencies as well as improved refractory on of steel will be reduced. The actual cost refractory makers are not charging double the materials. varies from 400-550 per ton, depending on price, at best 25 percent more than the cost

So, specific consumption of refractory in the logistics, steel grade produced and other of refractory made with magnesia carbon. But spinel is giving a performance that has steel is reducing every year. For instance, from factors. double life than magnesia carbon," sources at

a leading refractory maker said.

Under normal circumstances, the cost of spinel refractory is `70,000-80,000 per ton while magnesia carbon refractory pricing is pegged at `60,000 per ton.

#### Imports and exports

At present, in the total market of `7,000-crore, `2,500 crore comprises total imports. In this `2,500 crore pie, the share of China is almost 60 percent, say industry sources. Exports comprise around `1,200 crore of refractories, which makes India a net importer of refractories.

"China has a distinct advantage over us because of the availability of quality raw materials. The fuel is cheap, labour is also cheap. This country can afford to sell at a price which we cannot. That is why Chinese refractory makers put up a price that is abnormally low," says IRMA's Dasgupta.

But, it also has got to do with the Chinese steel market.



China's steel industry is producing 800-steel making. "Even if we hit 200 mt by 51 percent share and now it is the main 850 million tons of crude steel, of which 3002030, though as per the government policy, technology provider within the company. mt is surplus. Naturally, the manufacturers should be 300 mt, and even if refractory The top 5 refractory companies in India are dumping this excess material all across the consumption comes down from 11 kg to are bringing in technology from outside the world at throwaway prices without earning 8 kg per ton of crude steel, consumption country and these will be the players who any profit on it. Resultantly, refractory makers will amount to 200 mt multiplied by 8 kg. would look to set up plants within India. in China are also suffering and, in a backlash, So, there is opportunity for growth. If the However, Ali, Chairman, Imerys India move, are trying to explore new markets. And steel industry grows, we grow. But Indian and Managing Director, Calderys since India is the nearest market, they are refractory makers must be allowed to have a BU India & Middle East, expects the trying to dump their refractory material into share of the growth pie. If China takes away refractory market to stay more or less flat in the country, which is negatively impacting a good share of that market, then it will be the near term. He says: "India's refractory the domestic manufacturers of refractory problem."

"Sometimes the Chinese players quote such Sources at TRL Krosaki said, the Indian percent y-o-y since the last 5-6 years, mainly prices that are impossible to match up to government, irrespective of whoever is iron account of expansion in India's domestic considering our labour, overhead and other power, will have to push for infrastructure steel production. However, of late, the growth costs," said a source. development, without which there cannot may have slowed down to 4-5 percent owing

According to Ali, exports currently be fresh investment. And the government is to issues such as cheap imports, raw material accounts for ~15 percent of the total working in terms of rail, road development aspects and stringent government regulations refractory industry in India. Small refractory etc. because of increasing non-performing assets

players currently export majority of alumina- Thirdly, the per capital consumption (NPAs). based refractory and shaped refractory items of each and every commodity – steel, He adds: "Increase in stressed assets to the Middle East and East African nations. aluminium, copper, cement etc – in India, and NPAs and insolvency proceedings

India's potential for refractory compared to the world, and China average by the National Company Law Tribunal requirement is huge and only because of this is almost half. We have a high population (NCLT) against key steel players have current economic slowdown in India, local and hence in many cases the per capita impacted domestic steel production. As steel refractory suppliers are exporting, he adds. consumption formula doesn't work. In a accounts for 70 percent of the total refractory

"Refractory exports are usually a low-small country, the specific consumption will consumption, there has been a significant margin business and not very lucrative be very high because of the lower number impact on the refractory industry too. As However, local suppliers are currently of people living in that country. But that refractory follows steel trends, we expect the relying on exports to manage their capacity 15-25 percent minimum jump in per capita refractory market to remain more or less flat utilisation," adds Ali. consumption is required in India. in the near term."

Also, after the pollution issues are sorted "But, in view of the highly optimistic Growth prospects out in China, the refractory manufactured government initiatives, the refractory industry However, all hopes are not dashed. There is there is expected to become costlier and have expected to again grow at a rate of 6-7 silver lining glistening from behind the clouds lesser number of takers in China's domestic percent which we have seen till the couple of of China's pollution smog despite what the market and some of that material is also years back," stresses Ali, adding, "The current naysayers espouse. As per industry estimates, likely to come to India That cost advantage economic slowdown, due to factors such the market will grow at a 3.5 percent enjoyed so far by the manufacturers may as demonetisation, GST implementation compound annual growth rate (CAGR) for not be there anymore because of the strict issues, insolvency proceedings under NCLT the next 3 years in a scenario where actual benchmarks miners will have to follow in etc has impacted the refractory industry as consumption in India is currently at around terms of pollution and health. Many of the well. However we expect signs of recovery by 20 lakh tons per annum. refractory makers in India have plants in next fiscal."

As already mentioned, the iron and steel China, including TRL Krosaki, because Indeed, implementation of key industry is the largest user of refractory of the proximity to the raw material which government initiatives (Make in India, material, its share being 70-75 percent. It allows a cost advantage. But, post the Housing for All etc) should boost India today the installed capacity in iron environmental clampdown, producing in infrastructural growth and could alter the and steel is around 130 mt per annum with India may become cheaper. country's economic landscape. Growth the steel makers producing around 100 mt. Technology is not an issue with the Indian potential for India's refractories industry is Analysts Steel Insights spoke to say even if players. All the major refractory makers have thus expected to be in positive territory over they do not bring in new capacity, but only an overseas partner for technology back-up the longer term. Refractory consumption in utilise their existing capacity, there will be TRL, for instance, did not have any overseas the domestic market is bound to increase demand for refractory. partner but instead had technology transfer when economic activity resumes normalcy

As says Dasgupta of IRMA, around 11 agreements with some overseas players and core sectors starts performing up to kg of refractory is required per ton of crude However, in 2011, Krosaki Harima took expectations ■



# Refractory sector must focus on indigenous raw material R&D: IRMA

**T**he Indian refractory industry can grow if the steel industry grows. For every ton of crude steel manufactured, around 11 kg of refractory is required. In the government's 300 million tons of crude steel production plan by 2030, Indian refractory makers must be allowed to participate. If Chinese refractory makers take away a sizeable share, then it will pose a problem. Also, the stress should now be on developing R&D for indigenous raw materials. Working toward this end, the Indian Refractory

Makers Association (IRMA) has just signed a MoU with IIT-BHU, Varanasi for developing a refractory centre of excellence, Anirbandip Dasgupta, Senior Executive Officer, IRMA, tells Madhumita Mookerji.

We will explore what sort of reserves and deposits we have, or we do not have in India, what are the possibilities etc. We will be cataloguing this. This will happen in the first phase. In the second phase, we will have to see how we can develop these.

Already, some work has been done in relation to beneficiation but on a trial basis. It could not be scaled up to a commercial level. This is definitely a shortcoming of the refractory industry.

Secondly, we will have a third party laboratory on quality control. We do not yet have such a unit which gives reliable results.

Thirdly, since the Benares Hindu University (BHU) offers a course in ceramic engineering, we could encourage some of the ceramic engineering graduates to enter the refractory industry. This would address the human resource (HR) problem.

Fourth, we, at IRMA, organise training programmes for our members, which allows for re-skilling.

What role is IRMA playing and how can it further help the industry?

IRMA has just signed a memorandum of understanding (MoU) with IIT-BHU,

Varanasi to develop an excellence centre in this city. You spoke of raw material characterisation. Could you take the help of any organisation regarding this?

We will have a laboratory where we will do indigenous raw material characterisation. We have not really thought about this. If

needed, we can take the help of the Indian Bureau of Mines in mapping reserves or R&D. There are pitfalls in mapping as well.

For instance, we may map some extensive bauxite reserves but the actual reserves are often seen to be much less or filled with such impurities that it would not be economically viable to mine that block.

Does IRMA plan to take initiatives in beneficiation?

Yes, we do have plans. Because, unless we have our own indigenous sources of raw material supply we will continue to be buffeted by global ups and downs.

The Steel Development Fund (SDF) has also indicated it will support us, provided we put up a proper proposal. See, if we get raw material security, ultimately, the steel industry will only benefit.

Technology-wise, how advanced are Indian refractory makers at present?

Most of the Indian manufacturers have opted for tie-up with an overseas partner for technology access. Or there are overseas operators in India.

Thus, the technology is advanced. Vesuvius, Krosaki Harima, Calderys etc, to name a few, all have proven, cutting-edge technology. That is fine. But, what about developing indigenous technology based on indigenously available resources. And for that, we need talent.

Which are the areas in which we can stress on developing indigenous technology?

In raw materials, beneficiation, value-addition, developing of aggregates, on ways to cut down on energy costs.

There is also scope to increase the application areas. In steel-making, for instance, in blast furnaces, a lot of work has been done in the repair area. There is scope in ceramic welding.

There is scope for developing material that conserves greater energy in cement, and heat in sponge iron units, because conserving energy means money – decreasing the customer's cost of production.

There can be newer variants of magnesia

carbon, using additives for better life.

Through such means, we can thus give more value to the customer.

Has there been any impact on the industry since the rollout of GST?

Yes, there has been. The government has given us three rates – 5 percent, 18 percent and 28 percent. But we want one single rate to be applicable to this industry. We have now approached the GST council and are trying to resolve the issue.

Apart from raw material issues, are there any other challenges being faced by refractory makers?

Human resource is a major problem. Most feel it is a polluting and non-glamorous industry and avoid it. In the last 20 years, very few really good talent has come into this industry and our R&D is being impacted. Because, we need good research and development in refractory to offset this raw material issue.

We need out-of-the-box thinking and this is where India is facing a lag. In the last 20 years or so, all the good brains have gone to information technology, management etc rather than brick-and-mortar R&D.

Then there are payment issues. There

are payment clauses which create problems. If a player delivers his order today, he gets payment 6 months later and that too not the full amount. Capital gets blocked for almost a year.

Steel-makers habitually look to bring down the cost of refractory in a bid to cut production costs. Since they cannot touch power tariffs or iron ore costs, they always go for refractory prices!

So, where do you see the opportunities in the gloom?

Around 11 kg of refractory is required per ton of crude steel.

India is currently producing around 95 million tons (mt) of crude steel annually. As per the government policy we should produce 300 mt by 2030.

Even if we produce 200 mt and refractory consumption comes down from 11 kg to 8 kg per ton of crude steel, consumption will amount to 200 mt multiplied by 8 kg.

So, there is opportunity for growth. If the steel industry grows, we grow. But Indian refractory makers must be allowed to have a share of the pie. If China takes away a good share of that market, then it will be a problem. ■

# Calderys India may go south, east in the mid-term

**C**alderys India, a leader in monolithic refractories, serves industries ranging from iron and steel, cement and power to petrochemicals. However, with its main focus being on the steel industry, the latest developments plaguing the latter as well as the current macro-economic parameters are having a rub-off effect on the company too. However, Calderys India, looking into the future, is eyeing newer export geographies like Pakistan and putting more stress on the domestic cement industry where it already commands 40 percent share in monolithics, apart from upgrading its current facilities to bring in better energy efficiency and cost optimisation, **Hakimuddin Ali, Chairman, Imerys India and Managing Director, Calderys BU India & Middle East, informs Madhumita Mookerji.** Excerpts from an interview:

How has the performance of the steel industry over the last few years impacted Calderys' performance? What strategies has it adopted to survive the recession?

The iron and steel industry has been going through a tough phase since the last couple of years, owing to issues such as cheap imports, raw material aspects and stringent government regulations because of increasing non-performing assets (NPAs).

Increase in stressed assets and NPAs and insolvency proceedings by the National Company Law Tribunal (NCLT) against key steel players have impacted domestic

steel production. As steel accounts for 70 percent of the total refractory consumption, there has been significant impact on the refractory industry too.

Calderys' segmental growth in iron and steel has shown a marginal increase in calendar year (CY) 2016 compared to 2015.

Our niche product offerings such as shotcrete for torpedo ladles, robotic shotcrete repair for blast furnace stack, through management and focus on total management solutions will be key growth factors in the coming time.

What is Calderys' current turnover and growth targets?

Net revenues of Calderys India were ₹563 crore in financial year (FY) 2016-17. Our prime focus is the iron and steel segment with total management solutions in ladles, convertors, electric arc furnaces (EAFs), snorkels and RH degassers. Further, we aim to strengthen our position in the cement segment where we have more than 40 percent share in monolithics.

What is Calderys' product portfolio like and which are its bestsellers and why?

Calderys offers complete refractory solutions

based on monolithics, ready shapes and accessories, from conception to installation and caters to all industry segments.

We have a complete range of monolithics, high alumina and insulating bricks. Our bestseller monolithics are abrasion-resistant castables for the cement industry, gunning and shotcrete products, iron trough castables and the entire range of Whythead castables.

With our fully mechanised shotcrete solutions for the iron and steel industry (torpedo, blast furnace stack repair etc) and the cement industry, we have been adding value to customers in terms of faster installation rates (6-7 tons/hour), higher safety standards and minimum losses.

Calderys has a dedicated team for shotcreting and gunning and we are in a position to mobilise our manpower and equipment at a very short notice from clients.

Calderys contributes significantly to India's hot metal handling process through its trough management solutions. Major steel players such as the Steel Authority of India Limited (SAIL), Rashtriya Ispat Nigam Limited (RINL) and Tata Steel are being catered to with our trough management solutions. Customers are deriving value from our trough management solutions through reduced specific consumptions and efficient site management.

Where are Calderys' manufacturing facilities located and what are the installed capacities and capacity utilisation levels like?

We have 3 manufacturing units in India: These are located in Katni (Madhya Pradesh), Nagpur (Maharashtra) and Wankaner (Gujarat).

**Our niche product offerings such as shotcrete for torpedo ladles, robotic shotcrete repair for blast furnace stack, trough management and focus on total management solutions will be key growth factors in the coming time.**

The total installed capacity (of all the 3 plants) is approximately 220,000 tons with a capacity utilisation of around 70 percent.

What are the company's growth plans in terms of investment, expansion, new plants, etc?

Our short-term plan would mainly focus on upgrading our current facilities to bring in better energy efficiency and cost optimisation.

In the mid-term, we may assess the possibility of expanding our operations in southern/eastern India.

Our unique proposition has been our refractory service centres under our Calde Serve Mission where we provide immediate technical assistance to our customers from our nearest service centre point. We have been able to minimise customer equipment downtime through this innovative maintenance service, leading to higher productivity for customers.

Currently, we have 5 service centres located at Nagpur, Raipur, Katni, Wankaner and Guntur and have plans to set up a few more service centres in the vicinity of our customer base.

Any particular challenges being faced by the company at present? How can these be resolved?

One of the key challenges is the current macro-economic situation which has significantly impacted the growth of core sector industries. India reported one of the lowest GDP growths in recent times (5.7 percent in the first quarter or Q1 of 2017-18) and even the Index of Industrial Production (IIP) has revealed a significant slowdown trend.

Increase in stressed assets and NPAs has forced the government to take strict measures against loan defaulters, majority of which are steel players. The refractory industry too is facing the heat of the ongoing macro-economic situation.

Another key challenge is the uncertainty in raw material supplies. Price trends of many imported raw materials have shown steep increases, mainly on account of China's stricter environmental policies, leading to closure of most of the Chinese manufacturing units of refractory raw materials.

Even the domestic raw material supplies have shown uncertainty, as no new mining leases have been sanctioned in the last couple of years.

Finding alternate raw material sources apart from China and in-house R&D for substituting unavailable raw materials (RM) will be the key to overcome issues pertaining to these (raw materials).

What percentage of Calderys' production is consumed domestically and by exports? Which are its key export markets. Are any newer geographies being looked at?

Contribution from exports is around 10 percent. Calderys India's key export markets are primarily the Middle East and East Africa. We plan to serve the Pakistan market as well in the near future.

What is the share of steel and other industries in Calderys' client base?

All major integrated steel players such as SAIL, Tata Steel, JSW Steel, JSPL, BPSL etc are our key customers in the steel segment.

Apart from the integrated steel plants, majority of secondary mini steel mills producing alloy steels are also part of our customer base.

As far as the cement industry is concerned, major cement manufactures such as the Holcim Group, UltraTech Cement, the B K Birla Group etc are our key clients.

Apart from the iron and steel and cement industries, we also cater to almost all key players in the aluminum, power and petrochem industries.

# Moderate but continued growth expected for global steel demand: WSA



## Steel Insights Bureau

**T**he World Steel Association (worldsteel) has forecasted that global steel demand will reach 1,622.1 mt in 2017.

In 2018, it is forecast that global steel demand will touch 1,648.1 mt.

Worldsteel forecasted in its short range outlook for October 2017 that global steel demand, excluding China, will reach 856.4 mt, an increase of 2.6 percent in 2017 and 882.4 mt, in 2018, an increase of 3 percent.

Commenting on the outlook, T V Narendran, Chairman of the worldsteel Economics Committee, said, "Progress in the global steel market this year to date has been encouraging. We have seen the cyclical

upturn broadening and firming throughout the year, leading to better-than-expected performances for both developed and developing economies, although the MENA region and Turkey have been an exception."

The risks to the global economy that we referred to in our April 2017 outlook, such as rising populism/protectionism, US policy shifts, EU election uncertainties and China's deceleration, although remaining, have to some extent abated. This leads us to conclude that we now see the best balance of risks since the 2008 economic crisis. However,

escalating geopolitical tension in the Korean peninsula, China's debt problem and rising protectionism in many locations continue to remain risk factors, Worldsteel said.

"In 2018, we expect global growth to

moderate, mainly due to slower growth in China, while in the rest of the world, steel demand will continue to maintain its current momentum," the association said.

So, world steel demand is recovering well, driven largely by cyclical factors rather than structural. The lack of a strong growth engine to replace China and a long-term decline in steel intensity due to technological and environmental factors will continue to weigh on steel demand in the future.

Both advanced and developing economies are exhibiting stronger economic momentum this year. Confidence and investment sentiments are improving in large parts of the world despite some financial market volatility and growing concern of stock market overvaluation.

Also, on a positive note, global trade is gaining momentum despite worries about rising protectionism and talks of rearranging existing free trade agreements.

India had a slowdown in economic activity in 2017, but accelerating government reforms are expected to bring about a better investment environment, leading to growth in the coming years. Investment activities are still driven by government initiatives and private sector investment is still restrained due to leveraged corporate balance sheets.

The US economy continued to exhibit robust fundamentals supported by strong consumer spending and rising business confidence.

Japanese steel demand is showing better than expected performance benefiting from the government stimulus package, improving exports and preparations for the 2020 Olympic games. South Korea's steel demand is suffering from high consumer debts, weakening construction and a depressed shipbuilding sector, while escalated tension around the North Korean nuclear weapons threat poses a serious and highly unpredictable risk.

With these generally favourable developments, steel demand in the developed economies is expected to increase by 2.3 percent in 2017 and 0.9 percent in 2018.

The Chinese economy, which has been gradually decelerating, is increasingly supported by consumption while investment continues to decelerate. However, government stimuli, particularly a moderate boost to the construction programme, contributed to increased GDP growth in 2017.

China's steel demand is expected to increase by 3 percent in 2017, an upward revision over the previous forecast. Then the global economy. In Brazil, continuing recent closure of induction furnaces will lead to a one-off jump in measured steel use in 2017 to 12.4 percent. The outlook for China's steel demand in 2018 remains subdued, showing no growth over 2017 as the government resumes and strengthens its efforts on economic rebalancing and environmental protection, worldsteel said.

Countries in South America have been slow so far in benefiting from the recovery of the global economy. In Brazil, depressed construction activity has held demand recovery back in 2017 but a stronger use in 2017 to 12.4 percent. The outlook for China's steel demand in 2018 remains subdued, showing no growth over 2017 as the government resumes and strengthens its efforts on economic rebalancing and environmental protection, worldsteel said.

The construction sector in the developed economies, which had been slow to recover from its collapse after the 2008 economic crisis, is now showing more positive signs both in the residential and commercial sectors, due to rising incomes and improving investment sentiments. Infrastructure investment, which has been driving steel demand in developing countries, is likely to get some additional support from the developed world's infrastructure renewal initiatives.

Developing countries benefited from a strengthening global economy. The reform agendas in many developing countries such as Egypt, Brazil, Argentina, Mexico and India are expected to enhance their growth potential over time.

ASEAN remains a high growth region, especially Vietnam and the Philippines, while more mature economies such as Thailand and Malaysia, are showing slower growth.

In the CIS, steel demand is expected to strengthen in 2017-2018 and specifically Russia is likely to maintain its slow recovery.

Turkish steel demand is expected to resume growth momentum in 2018.

The MENA region's outlook has suffered from low oil prices, geopolitical strife and high inflation. The region would benefit from reconstruction efforts once the major conflicts are ended. GCC countries continue to struggle with the low oil price environment.

The board elected Kosei Shindo, Representative Director and President, Nippon Steel and Sumitomo Metal Corporation, as the Chairman, the body said.

John Ferriola, Chairman, Chief Executive Officer, President, Nucor Corporation, and POSCO CEO Ohjoon Kwon have been elected to the post of Vice Chairman.

Chairman and Managing Director of Indian steel giant JSW Steel, Sajjan Jindal, has been elected as the Treasurer of worldsteel.

The board also elected a 16-member Executive Committee. Jindal and Kwon were also elected as one of the members of the committee. Narendran and NRI steel baron Lakshmi Niwas Mittal were also elected as members.

The global automotive sector is reporting a strong performance in 2017 with an especially strong performance in Turkey and Mexico. However, in the US and China, the auto sector could moderate and this trend is likely to extend to other countries in 2018, the association said in its short range outlook.

Sajjan Jindal, T V Narendran elected to key worldsteel posts

JSW Steel Chairman-cum-Managing Director Sajjan Jindal, has been elected



Besides, John Ferriola of Nucor Corporation, Heinrich Hiesinger of thyssenkrupp AG, André Johannpeter from Gerdau SA and JFE Steel Corporation's Koji Kakigi were elected to the committee.

M A Guoqiang of China Baowu Steel Group Corporation Limited, Alexey Mordashov of Severstal (PAO), Roger Newport of AK Steel Corporation, Paolo Rocca of Techint Group, Kosei Shindo of Nippon Steel and Sumitomo Metal Corporation, YU Yong of HBIS Group Co and World Steel Association's Edwin Basson were also elected as a member of the executive committee.

The World Steel Association (worldsteel) is one of the largest and most dynamic industry associations in the world.

Worldsteel members represent approximately 85 percent of the world's steel production, including over 160 steel producers with 9 of the 10 largest steel companies, national and regional steel industry associations, and steel research institutes. ■

# Coking coal offers ease in this October, may remain soft in winter



Arindam Bandyopadhyay

Seaborne coking coal prices showed a decline in October 2017, mainly due to low procurement by China and India, amid expectations that prices may ease further in the winter months, industry sources said. Also, Chinese end users of seaborne coking coal have stopped making purchases ahead of measures to cut emissions, sources said. “Some traders are still in the market with the hope that the impact of steel production cuts in China would not be too pronounced and demand will continue to hold in the coming weeks,” a source said.

According to information available with Steel Insights, the premium variety was quoted lower at \$180 per ton FOB Australia on October 30, compared to \$189 per ton FOB Australia as on September 29. Peak prices were quoted at \$181 per ton FOB Australia on October 30 as compared to \$190 per ton FOB Australia on September 29. There was also a drop in semisoft prices during the month. Some traders said they have no need for spot coking coal cargoes, and expect prices for premium cargoes to drop. In the absence of Chinese demand, seaborne coking coal sellers are targeting the Indian market where there were some buyers to procure spot cargoes. Another steel market source said steel markets had been performing “reasonably well and some Indian buyers are keen to conclude spot deals.”

The sources said the decline in seaborne coking coal prices was on the expected line in the absence of fresh offers and bids. Although Australia-origin seaborne cargoes remained on offer in the market there was less number of takers in the market. Meanwhile, India's hot metal production in September 2017, at 5.576 million tons (mt), was up by 0.3 percent over August 2017 and up by 4.7 percent over September 2016. During April-September 2017, hot metal production was 32.385 mt, a growth of 0.9 percent over same period of last year, provisional steel ministry data showed.

“We are not taking spot cargoes at the moment due to the long wait time at the ports and because our term contract tonnages are enough for our current needs,” a trading source in India said. This indicates stable demand for coking coal from Indian steel mills, looking for a sustained supplies of the steel making raw material, which is prone to intense volatility in the recent months owing to varied factors like bad weather and disruptive evacuation in Australia. India's 12 major government-owned ports handled 24.78 mt of coking coal during April-September, a 3.31 percent rise as compared to around 23.99 mt recorded for the corresponding period a year ago, according to the latest data released by the Indian Ports Association.

India has to heavily depend on import of coking coal, as the domestic quality has higher ash content and not suitable for steel industry with present technology. Country's present coking coal production is around 50 mt out of which only 4-5 mt is being used by the steel industry, and the major chunk goes to thermal plants.

In the near term, the spot prices in the global markets are expected to remain soft, if supply scenario remains unaffected. There is a downtrend in China's domestic met coal and coke market and the prices are expected to fall further. As a result, the sentiment in the global market is also expected to remain bearish, the sources said.

Met coke import prices down Meanwhile, after a surge in September, metallurgical coke import prices for India slid in the month of October 2017, tracking the decline in coking coal prices in the international markets, according to Steel Insights.

Imported met coke prices slid to \$334 per ton CFR India on October 30, 2017, from \$383 per ton CFR India on September 29, 2017.

The movements in met coke prices generally comes with a lag following movements in spot coking coal prices, market sources said, adding that the expectation is that the prices would soften further as coking coal prices slide in coming months.

In the domestic market, met coke prices remained unmoved in October 2017. According to sources, domestic met coke prices stood at around \$22,500 per ton (ex-plant) in the east coast as of October 30, 2017, compared to same prices in September 2017. Prices of met coke (ex-plant) was at \$29,500 per ton (ex-plant) in the west coast on October 30, 2017, against same prices in a September 30, 2017.

# Pellet makers seek total ban on high grade iron ore exports

India's iron ore exports  
(in million tons)

| Year | Exports |
|------|---------|
|      |         |
|      |         |
|      |         |
|      |         |
|      |         |
|      |         |

Source: Insights Research



Leading pellet manufacturers also feel ban on high grade iron ore fines exports would increase its availability in the domestic market as such ore is in high demand by steel units.

“In any case, exports on raw material should never be encouraged since the value to the same resource can be increased exponentially by value addition which does not cost as much. Even smaller countries in Africa do not allow export of raw material like iron ore and have laws which encourage industrialization and export of finished products,” pellet makers said.

Unlike top iron ore exporting nations like Australia and Brazil, India has a large steelmaking capacity that needs to be backed by ample iron ore supplies. India is the third largest steel producer with a rated capacity of 125 million tons (mt). Actual steel output in the country in FY17 was 97.4 mt, consuming 155.78 mt of iron ore against a total ore production of 190 mt. The country’s iron ore exports jumped four fold to 24.35 mt in FY17 on strong global prices.

By 2030-31 when the country targets steel capacity of 300 mt, iron ore requirement is projected to climb to 437 mt.

The domestic pellet industry tasked with value addition has a capacity of 85 mt created with an investment of 40,000 crore. However, actual output was only 48.5 mt in last fiscal.

Capacity utilisation (of pellet plants) can be significantly increased if the raw material is available at competitive price. Domestic market is oversupplied with lump and pellet since the steel makers are not able to produce at their installed capacity for multifaceted reasons including the market challenges.

Export of pellets should help the pellet manufacturers to stabilize their operation, attain rated capacity and thus reduce per ton cost of manufacturing, sources said.

Steel Insights Bureau

Concerned over rising trend in iron ore exports, pellet manufacturers have called for a total ban on exports of high grade iron ore fines from the country in the interest of the local steel industry.

“High grade iron ore is a scarce resource that needs to be conserved for value addition,” they said.

Pellet makers who are forced to operate at depleted capacity, are also pitching for 30 percent export tax on lower grade fines (iron content of 45 to 58 percent). The Union government has exempted export duty for low grade iron ore fines (iron content below 58 per cent) while continuing with the 30 per cent duty on high grade iron ore.

Federation of Indian Mineral Industries (Fimi) has been lobbying for a cut in export duty on higher grade iron ore. To study the possibility of rationalisation in export duty,

the Union mines ministry has formed a committee.

But, Pellet Manufacturers Association of India (PMAI) is opposed to any move to prune or abolish export duty on iron ore.

Iron ore is national wealth and needs to be preserved for the future especially when the country is eyeing 300 million tons steel capacity (by 2030-31). If there is glut of low grade iron ore fines, the same can be made available to pellet plants at a nominal cost.

It was felt that the cut off for export duty should be iron ore fines with 45 percent instead of 58 percent grade iron ore, said sources at PMAI.

Justifying the export duty demand for 45-58 percent grade fines, they said, Institute for Minerals and Materials Technology Export of pellets should help the pellet manufacturers to stabilize their operation, attain rated capacity and thus reduce per ton cost of manufacturing, sources said.

IMMT, Bhubaneswar is working on new technology that can beneficiate even 45 percent grade iron ore fines.

# Railways tender an opportunity for steel mill

Steel Insights Bureau

Indian Railways has floated a global tender to procure rails and the state-run freight and passenger carrier is seeking to buy 700,000 tons of rails for track upgrades, Railway Minister Piyush Goyal said.

Analysts say apart from SAIL, the move will ensure steady supplies and competitive prices and benefit Jindal Steel & Power Ltd.

Prime Minister Narendra Modi is pouring about 8.6 trillion to upgrade the aging lines of the network that was started under British colonial rule. Giant track-laying projects are underway to modernize passenger and freight movements to cut travel time.

The tender gains significance as Steel Authority and Jindal Steel have been trying to boost market share and profits after reporting losses for at least two straight years even as they battle high debt levels. Indian Railways expects annual demand for steel rails to rise to 1.5 million tons in the year ending March from about 800,000 tons in the prior 12-month period.

Recently, Steel Minister Birender Singh asked SAIL to ramp up production from its new rail mill in Bhilai to meet the requirements of its major customers. “The company should chalk out a time-bound action plan to cater to the Indian Railways’ increasing demand.”

Expecting demand to surge from railways to housing, steelmakers have lined up expansion plans. Annual capacity stands at 125-126 million tons and is forecast to rise to 150 million tons by 2021, before settling at 300 million tons, according to Steel Secretary Aruna Sharma.

Track renewal and safety is one of the most important measures being implemented by the railway ministry in the backdrop of several train derailments that the national carrier has witnessed.

“So, seven lakh metric ton of additional rail (track) is sought to be procured for which

a global tender is already been out on the 12th of October,” Goyal said.

Currently, state-run Steel Authority of India Ltd (SAIL) supplies tracks to the national carrier.

The additional rail tracks will help the railways towards clearing the track renewal backlog. The people’s carrier has a track length of around 115,000km, making it the world’s largest railway network under a single management. Of this, the route length—the length of track not including that at yards and railway stations—accounts for 67,312km.

Indian Railways is the country’s largest employer with 1.4 million personnel. It runs around 20,849 trains daily and transports 23 million passengers and three million tons of freight. It operates 10,773 locomotives, 63,046 coaches and 245,000 wagons.

| India's steel consumption by railways (in million tons) |                              |
|---------------------------------------------------------|------------------------------|
| Year                                                    | Railway material consumption |
|                                                         |                              |
|                                                         |                              |
|                                                         |                              |
|                                                         |                              |
|                                                         |                              |
|                                                         |                              |
|                                                         |                              |
| \$ S U 6 H S W                                          |                              |

Railways: a major stainless steel consumer  
The Indian Railways is the largest railway system in the world under a single ownership. The Indian Railways is drawing up an ambitious modernization plan spread over the next five years. This amount will be spent on modernization of railway signaling system, using all-stainless steel coaches, track and rolling stock so that modern services could be provided to passengers in their coaches and platforms and freight customers at low prices by bringing down the unit cost.



The Indian Railways have been producing painted stainless steel passenger coaches for important trains such as Rajdhani and Shatabdhi. Now the railways plan to convert all new production of railway passenger coaches (about 3,000 per annum) to unpainted stainless steel coaches. The Railways are considering 300 series (nickel containing) stainless steels for all exterior surfaces on account of their superior corrosion resistance to withstand the harsh tropical climate in India and a suitable cost-effective steel for all internal structural members.

409M stainless steel is being tendered by the Indian Railways to make coal wagons. Indian Railways is considering stainless steel options for wagons for ores, minerals, fertilizers, food grains, lime stone, fly ash, cement, oil tankers. Expansion of Delhi Metro’s services and adoption of the Delhi Metro-type transport system, using all-stainless steel coaches, is being replicated in other cities like Mumbai, Bangalore, Hyderabad, Ahmedabad, Chennai. This will significantly add up to stainless steel tonnage. Each coach uses 12 tons of stainless steel.

# Kolkata Port Trust plans 3 terminals at Haldia



Steel Insights Bureau

**T**he Kolkata Port Trust (KoPT) plans to augment capacity at Haldia port by adding three new terminals at an estimated cost of ₹800 crore.

A total of 9 million tons will be added to the existing handling capacity of 43 million tons at Haldia.

While one liquid-cargo terminal (2 million tons), primarily for edible oil, has been proposed at Salukhali, 15 km north of Haldia; two outer terminals — one for dry-bulk cargo (5 million tons) and another 2 million-ton liquid-cargo terminal — have also been planned.

Work on all the three projects is expected to begin next fiscal once environmental clearances are received.

According to Vinit Kumar, Chairman, KoPT, all three terminals will be developed on the public-private partnership (PPP) model.

While around ₹200 crore will be spent towards developing the Salukhali terminal, the liquid-cargo terminal, called Outer Terminal-II, is expected to entail an

outgo of ₹100 crore. The remaining ₹500 crore will be spent on setting up Outer Terminal-I.

“Haldia is a riverine port, and by having these projects on the outskirts, larger vessels can be accommodated. We will also gain with the better draft at these terminals,”

As against a 7-metre draft at Haldia, the proposed Salukhali terminal, for example, will have at least a 9-metre draft.

The Haldia Dock Complex handled around 19.124 million tons of cargo between April and September this year, a near-18 percent rise over the 16.242 million tons handled in the year-ago period.

These projects apart, KoPT will also be investing ₹100 crore for laying a second rail line from Durgachak to Haldia. This will help speed up cargo loading. The South Eastern Railway is the implementing agency for the project, Kumar said.

The ₹50-crore project to augment track capacity is underway at the container-jetty and taking the help of the Inland Waterways Authority in ensuring movement of containers in and out of the city, avoiding the city roads.

a vessel and loading them for the onward journey to the rake, from the prevailing seven days, to 4.5 days.

The initiative might help the Kolkata port regain the confidence of Nepalese importers who are now testing more efficient cargo-handling practices at the Visakhapatnam port.

Kumar, however, pointed out that delay on the part of CONCOR (Container Corporation of India) in supplying rakes is also a reason behind the slow movement of Nepal-bound cargo from the port. “Cargo loading for Nepal has stopped for the last two days due to unavailability of rakes from CONCOR,” he said.

The port is also investing in parking lots to improve the loading rate. Currently, restrictions on truck movement imposed by the Kolkata police are affecting loading.

Kumar is also exploring the possibility of investing in a Ro-Ro (roll-on/roll-off)

# India offers golden opportunities for global investors: Gadkari

Steel Insights Bureau

India offers “golden opportunities” for global investors in sectors such as infrastructure, transport, agriculture and food processing, Union Shipping minister Nitin Gadkari said.

Not only is the country going to offer a world class infrastructure in highways in two years, waterways and agriculture are also undergoing massive reforms that include 325 irrigation projects and massive river-interlinking programme, the Road, Transport, Highways, Shipping, Water Resources, River Development and Ganga Rejuvenation, minister said.

Addressing the World Food India 2017 conference on the ‘Opportunities in Infrastructure Technology & Equipment’, Gadkari said, “There are golden opportunities for investors in food processing, agriculture, water transport, highways, shipping and other sectors.”

“We are bound to create a world-class infrastructure in coming two years. Of the 32 logistic parks, 24 have already been identified on National Corridors to be built at a cost of Rs 2 lakh crore. These parks will cater to key production and consumption centres accounting for 45 per cent of India’s road freight,” the minister said.

He said the work has already been started at Chennai, Bangalore, Vijaywada, Hyderabad, Surat and Guwahati for logistic parks that are designed to house cold storages and warehouses.

Besides, government’s another major and Mumbai-Kanyakumari will result in

initiative Sagarmala for port-led economic development of the country will see reduction of about `40,000 crore logistic cost in the country, he said.

Sagarmala will see development of 14 coastal economic zones besides development of two mega food processing parks, he said.

The minister stressed that development of logistic parks is essential as India witnesses wastage of fruits and vegetables worth 13,300 crore which account for about 25 percent of the produce due to lack of infrastructure.

Unfortunately, he said only 10 percent of fruit and vegetables are processed in India whose participation is barely 1 per cent in global processed food.

The government aims at enhancing the processing of fruits and vegetables to 20 per cent by 2019, he said.

He said construction of 44 economic corridors that include Mumbai-Kolkata

quick delivery of produce to markets besides reduction in logistic costs and enhancement in farmers’ income.

Talking about highways, he said that Eastern Peripheral Expressway to decongest Delhi will be made operational by next month while electronic tolling on highways pan-India will result in seamless flow of traffic throughout the country and would contribute in reduction of logistic cost which is at 18 percent in India as comparison to a mere 8 to 10 percent in China.

On irrigation front, he said, “As many as 325 new projects are being taken up by the government to help state governments in canal and other systems.”

Besides, he said, there is huge emphasis on inter-river-connectivity projects and 30 such projects will be taken up a cost of lakh crore.

“Out of these 30 projects, three important projects totaling an investment of about lakh crore will be taken up in three months’ time,” the minister said.



# Government bets on infrastructure to revive economy

Steel Insights Bureau

**T**he NDA government seems to be placing all its bets on infrastructure to revive the sagging economy.

Last month, it decided to invest a massive ₹7 lakh crore (including ₹1-2 lakh crore private investment) to develop over 66,000 kms of highways under the ambitious Bharatmala programme.

The aggressive push to increase public expenditure only confirms the intent to aid growth. It's known that infrastructure

investment creates jobs, boosts growth and, according to industry analysts. One, crowds in non-corporate private investment, almost all road projects are implemented which fell to a historic low and is refusing under the public-private partnership mode, to rebound. As per estimates, an increase in public infrastructure investment by 5 percent have lost their appetite. Many private of GDP increases growth by about 1-1.2 infrastructure firms are burdened with debt per cent, which in the current circumstances and out of luck to commit fresh investments. appears highly essential. Growth has been slowing down for the past six quarters to debt just to stay afloat. In this backdrop, how touch an 3-year low of 5.7 per cent for the incoming projects will be lapped up is first quarter ended June, 2017. unclear.

The proposed investments can stimulate the economy, but its success relies on two key elements, which seem unfavourable right now, according to industry analysts. One, crowds in non-corporate private investment, almost all road projects are implemented which fell to a historic low and is refusing under the public-private partnership mode, to rebound. As per estimates, an increase in public infrastructure investment by 5 percent have lost their appetite. Many private of GDP increases growth by about 1-1.2 infrastructure firms are burdened with debt per cent, which in the current circumstances and out of luck to commit fresh investments. appears highly essential. Growth has been slowing down for the past six quarters to debt just to stay afloat. In this backdrop, how touch an 3-year low of 5.7 per cent for the incoming projects will be lapped up is first quarter ended June, 2017. unclear.

leaves much to be desired. It's baffling that implementation, has set up standing some of the government projects have been committees fixing responsibility for time stuck for over 40 years at a stretch, either and cost overruns. But despite constant for want of approvals or finance. Take the efforts, project implementation is behind Howrah-Amta-Champadanga railway target. For instance, during April-June 2017, project in West Bengal initiated way back in the NHAI worked on 670 kms of national 1974 or the multi-state Nangaldam-Talwara highways, which was 42 per cent lower than rail line proposed in 1981 that are yet to be targeted 1,150 kms.

see the light of the day. As on today, costs But the upshot is that there's a significant doubled and trebled respectively for both improvement when compared to last two projects, but they remain non-starters. decades. An analysis of cost overruns in the In fact, as on July 2017, of the 1,257 last 18 years with respect to the originally central sector projects with an anticipated approved costs shows that the cost overruns project cost of 17 lakh crore, nearly half or reduced from 36 per cent in March, 2000 to 668 projects have inadequate information 11 per cent as on July, 2017. Likewise, the about their commissioning schedule, time over runs in the last 18 years fell from 45 per cent in March, 2000 to 22 per cent in July, 2017. This shows that various policies and control measures undertaken by the government seem to be doing delivering.

A mere 11 projects are ahead of schedule, and 304 on schedule, while 274 projects are delayed and 331 have cost overruns. Of the 1,257 projects, road transport and highways account as the highest number of projects at 480. Of this 63 projects are delayed and 42 projects have cost overruns. Similarly, railways — the second highest sector at 350 projects — a whopping 202 projects have cost overruns, while 33 are behind schedule. The other major sectors include petroleum and power, which together have 226 projects.

Some of the reasons for time over runs include delay in land acquisition, obtaining forest and environment clearances, lack of infrastructure support and linkages, delay in tie-up of project financing, changes in scope etc. Causes of cost escalation include under estimation of original cost, fluctuations in forex rates and statutory duties, spiralling land acquisition costs and general price rise.

In order to reduce delays and prevent cost overruns, the government initiated several measures including an online monitoring system to track status of project

India's long steel consumption pattern (in million tons)

| Year | long product consumption |
|------|--------------------------|
|      |                          |
|      |                          |
|      |                          |
|      |                          |
|      |                          |
|      |                          |
|      |                          |
|      |                          |
|      |                          |

Source: Insights Research

2017-18, as against 20.775 million tons in April-September, 2016-17, the data showed. Consumption of bars and rods rose 1.3 percent to 16.525 million tons during the period, while consumption of structurals fell 2.1 percent to 3.851 million tons during April-September, 2017-18. Railway material consumption rose 5.3 percent to 561,000 tons during the period.

It is estimated that construction and infrastructure constitute around 61 percent of steel consumption followed by capital goods (11 percent), automobile (8 percent), packaging (5 percent), consumer durable (3 percent) and others (12 percent).

Steel's high strength and relatively low cost accounts for valued use in many industries and provides the necessary raw material for extensive industrial supply chains. It provides the foundation for construction (bridges, buildings), transportation systems (railroads, cars, trucks, ships), utility systems (municipal water systems, power systems), as well as other diverse applications including military equipment, food storage, appliances and tools. Iron is not used directly because of its softness, but is combined with carbon, which then results into a tougher and more usable material – steel.

Rise in infrastructure development and automotive production would drive growth in the sector. Power and cement industries are also aiding growth in the metals and mining sector. Demand for iron and steel is set to continue, given the rapid growth expectations for the residential and commercial building industry.

India holds a fair advantage in cost of production and conversion costs in steel and alumina. Its strategic location enables convenient exports to developed as well as the fast-developing Asian markets.

Long product production for sale & consumption (in '000 tons)

| Product                         | Production for sale     | Consumption             |                         |        |
|---------------------------------|-------------------------|-------------------------|-------------------------|--------|
|                                 | April-September 2017-18 | April-September 2017-18 | April-September 2016-17 | Growth |
|                                 | Non-alloy               | Non-alloy               | Non-alloy               | %      |
| % D U V U R G V                 |                         |                         |                         |        |
| Structurals                     |                         |                         |                         |        |
| 5 D L O Z D \ O D W H U L D O V |                         |                         |                         |        |
| 7 R W D O                       |                         |                         |                         |        |

# India imposes anti-dumping duty on stainless steel products

Steel Insights Bureau

In a bid to protect local steel manufacturers from cheap imports, the Indian government has imposed anti-dumping duty on certain stainless steel products from China, the United States, South Korea and European Union.

The government in a notification said that the duty will be in effect until December 10, 2020 while exempting certain grades of stainless steel from the levy.

The decision to impose anti-dumping duty has been taken by the revenue department following recommendation by the Directorate General of Anti-Dumping and Allied Duties.

As per a notification issued by the department, the duty ranges between 4.58 percent and 57.39 percent of the landed value of cold-rolled flat products of stainless steel. The anti-dumping duty will be imposed on the imports of stainless steel products from China, Taiwan (China Taipei), South Korea, South Africa, Thailand, the United States of America, and the European Union.

Duty imposed on certain Chinese, EU steel products

Earlier this month, the government imposed an anti-dumping duty on the import of some flat steel products from China and the European Union for five years. Countries initiate anti-dumping probes to determine if the domestic industry has been hurt by a surge in below-cost imports. As a counter-measure, they impose duties under the multilateral WTO regime.

The duty was imposed after the commerce ministry's directorate general of anti-dumping and allied duties (DGAD) recommended duty on such imports.

In its findings, the DGAD had concluded

that 'colour coated/pre-painted flat products of alloy or non-alloy steel' has been exported to India from these regions at below normal value, due to which domestic industry has suffered material injury.

"The anti-dumping duty imposed under this notification shall be effective for a period of five years (unless revoked, superseded or amended earlier) from...11th January and shall be payable in Indian currency," the department of revenue said in a notification. The duty will be the difference between the landed value of the steel products and US\$ 822 per ton.

India has already slapped anti-dumping duty on certain cold-rolled flat steel products from four nations, including China and South Korea.

While DGAD recommends the duty to be levied, the finance ministry imposes it.

Countries initiate anti-dumping probes to determine if the domestic industry has been hurt by a surge in below-cost imports. As a counter-measure, they impose duties under the multilateral WTO regime.

Anti-dumping measures are taken to ensure fair trade and provide a level-playing

field to the domestic industry. They are not a measure to restrict imports or cause an unjustified increase in cost of products.

US slaps anti-dumping duties on steel rods from 7 countries

Meanwhile, the US has announced anti-dumping measures on carbon and alloy steel wire rods from Italy, South Korea, South Africa, Spain, Turkey and the Ukraine and the United Kingdom imposing duties of up to 147.6 percent.

Washington held that the countries sold their goods in the US at "less than fair value," according to a statement.

The duties, which differed for various companies within the named countries, ranged from 2.8 to 147.6 percent.

The measures are preliminary until the conclusion of two separate investigations on January 11, 2018, one by the Department of Commerce and the other by the US International Trade Commission (ITC) which was established to provide impartial, independent verdicts.

In 2016, imports of wire rod from Italy, Korea, South Africa, Spain, Turkey, Ukraine, and the United Kingdom were valued at an estimated US\$ 12.2 million, US\$ 45.6 million, US\$ 7.1 million, US\$ 40.7 million, US\$ 41.4 million, US\$ 55 million, and US\$ 0.5 million, respectively.

The measures came about following petitions filed on behalf of Gerdau Ameristeel, Nucor Corporation, Keystone Consolidated Industries, and Charter Steel.

Similar measures have already been taken against steel wire imports from Belarus, Russia and the United Arab Emirates.





## Car sales hit speed-breaker in October

Ritwik Sinha

Vehicle sales in October 2017 showed a downtrend with most car-makers to 32,490 units during the month under highest retail of over 50,000 units in a festival showing negative topline. During the review, from 33,929 in October 2016, the month, with wholesales of 49,588 units on month, Maruti Suzuki India Limited (MSIL), automaker said. The auto major further said the strength of strong demand for the Grand Toyota and Tata were the only 3 companies sales of the compact segment, comprising the i10, Elite i20 and Creta." Commenting on the October sales, Rakesh Srivastava, Director, Sales and Marketing, HMIL, said, "Hyundai did its highest retail of over 50,000 units in a festival showing negative topline. During the review, from 33,929 in October 2016, the month, with wholesales of 49,588 units on month, Maruti Suzuki India Limited (MSIL), automaker said. The auto major further said the strength of strong demand for the Grand Toyota and Tata were the only 3 companies sales of the compact segment, comprising the i10, Elite i20 and Creta." Home-grown auto major Tata Motors' October sales remained almost flat (marginally up by 0.72 percent) at 53,197 units (including exports) compared to 52,813 in the corresponding month of last year. The Indian arm of South Korean automotive manufacturer Hyundai Motor units clocked in the corresponding month of October 2017, growing 9.5 percent over India registered domestic sales of 49,588 units in the same period of the last fiscal. MSI's total units for October 2017 which was marginally lower by 0.85 percent compared to 50,016 units sold in the same month a year ago. The company has yet not given out the export numbers for the month. MSI said in a statement. However, sales of mini segment cars, including the Alto and WagonR, witnessed a 4.2 percent decline. Commenting on the October sales, Rakesh Srivastava, Director, Sales and Marketing, HMIL, said, "Hyundai did its highest retail of over 50,000 units in a festival showing negative topline. During the review, from 33,929 in October 2016, the month, with wholesales of 49,588 units on month, Maruti Suzuki India Limited (MSIL), automaker said. The auto major further said the strength of strong demand for the Grand Toyota and Tata were the only 3 companies sales of the compact segment, comprising the i10, Elite i20 and Creta." Home-grown auto major Tata Motors' October sales remained almost flat (marginally up by 0.72 percent) at 53,197 units (including exports) compared to 52,813 in the corresponding month of last year. The Indian arm of South Korean automotive manufacturer Hyundai Motor units clocked in the corresponding month of October 2017, growing 9.5 percent over India registered domestic sales of 49,588 units in the same period of the last fiscal. MSI's total units for October 2017 which was marginally lower by 0.85 percent compared to 50,016 units sold in the same month a year ago. The company has yet not given out the export numbers for the month. MSI said in a statement. However, sales of mini segment cars, including the Alto and WagonR, witnessed a 4.2 percent decline.

growing demand for newly-launched products and higher customer uptick owing to increased consumption across sectors coinciding with peak festive season buying trends across segments. Cumulative sales growth of commercial units in the domestic market for the fiscal (April-October, 2017) were at 193,782 units, a growth of 5 percent compared to 184,871 units, in the same period, last fiscal.

The Indian subsidiary of the American vehicle segment, M&M sold 748 units for the month, a growth of 59 percent. Exports units recorded sales of 16,475 units, growth of 1 percent, over October 2016. This was supported by continued good performance of new generation units Tiago, Tigor, Hexa and an excellent response received for the newly-launched compact SUV, Nexon. Cumulative sales growth of passenger units in the domestic market for the fiscal (April-October, 2017) were at 97,892 units, a growth of 10 percent, compared to 88,976 units, in the same period last fiscal.

The company's sales from exports were at 4,311 units in October 2017, a decline of 32 percent, due to continued drop in its ramp-up for new model introduction has impacted our wholesale for October. We are excited to have the all-new Ford EcoSport very soon."

Meanwhile, Japanese automaker Toyota Kirloskar Motors registered a total growth of 6 percent in October 2017 at 14,000 units compared to the 12,625 units from October 2016. Toyota Kirloskar Motor sold a total of 12,403 units in the domestic market compared to 11,651 units sold in October 2016. The company exported 1,597 units of the Etios series in October 2017 compared to the 974 units exported last year during the same month.

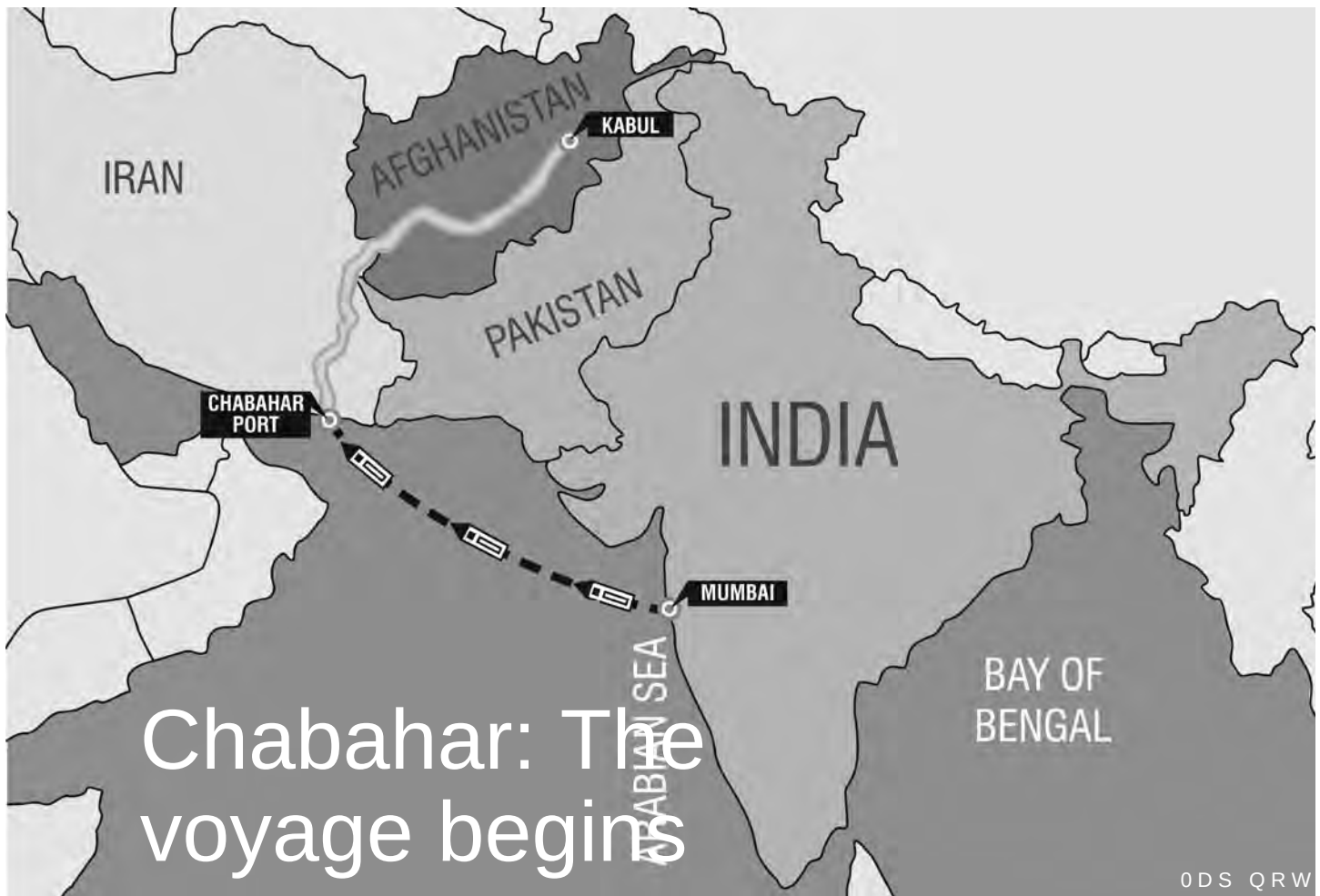
Commenting on the monthly sales, N Raja, Director and Senior Vice President, Sales & Marketing, Toyota Kirloskar Motor,

sales went up 7 percent as the company sold 19,279 units in October 2017 compared to 18,059 sold during the same month last year. Exports took a major hit last month at 2,331 units against 3,279 exported in October 2016, seeing a drop of 29 percent. Last month, M&M also launched the facelifted Mahindra KUV100 in India. In the medium and heavy commercial segment, the month, a growth of 59 percent. Exports for October 2017 stood at 2,331 units, a decline of 29 percent. Commenting on the monthly performance, Rajan Wadhera, President, Automotive Sector, M&M, said: "The auto industry has had a mixed month. The build-up to Dhanteras and Diwali was good but the demand tapered off subsequently. Last year, all festivals were in the month of October and the industry volumes had a high base. This year, September and October combined, we have grown by 7.9 percent in auto division." Managing Director, Ford India, said, "The Honda Cars India (HCIL), the passenger cars manufacturer in India, registered monthly domestic sales of 14,234 units in October 2017, against 15,567 units in the corresponding month last year. The company has registered a cumulative growth of 17 percent, selling 105,503 units during April-October, 2017, against 90,422 units in the corresponding period of April-October, 2016. The passenger vehicles segment, which includes utility vehicles (UVs), cars and vans, sold 23,413 units in October 2017, down 5 percent against 24,737 units during October 2016. The company's domestic sales stood at 48,818 units during October 2017 against 48,729 units during October 2016. The festive purchases also spread across 2 months this year, where we have witnessed a combined sales growth of over 6 percent."

Sales of top 7 Indian auto makers during October 2017 and 2016

| Company                 | October 2017 (in units) | October 2016 (in units) | Growth percent |
|-------------------------|-------------------------|-------------------------|----------------|
| 0DUXWL 6X]XNL ,QGLD     |                         |                         |                |
| + \XQGDL 0RWRU ,QGLD    |                         |                         |                |
| 7DWD 0RWRUV             |                         |                         |                |
| 7R\ RWD .LUORVNDU 0RWRU |                         |                         |                |
| )RUG ,QGLD              |                         |                         |                |
| 0DKLQGUD 0DKLQGUD       |                         |                         |                |
| +RQGD &DUV ,QGLD        |                         |                         |                |

India's H1 non-alloy flat steel consumption up 8.3%  
India's non-alloy flat steel consumption rose 8.3 percent in April-September, 2017-18 to 16.85 mt on higher procurement by end-users, provisional steel ministry data showed.  
Consumption of HR coils rose 16.9 percent to 7.52 mt during the period, while consumption of plates rose 0.4 percent to 2.45 mt during H1 FY18.  
GP/GC sheet consumption rose 8.5 percent to 3.65 mt, while CR coils/sheet consumption fell 3.7 percent to 1.89 mt.



Kingshuk Banerjee

**A**t last the ball has been set rolling for Chabahar.

While the recent flagging off of vessels carrying wheat from India to the Iranian port of Chabahar is definitely positive tidings as far as Indian initiatives in this part of Asia are concerned, the railway link still remains a dream. Therefore, unloaded cargo from Chabahar will have to go all the way to Afghanistan using the road network.

Meanwhile, two terminals at Chabahar could be fully operational by the end of 2018, Arun Gupta, Managing Director of India Ports Global Private Ltd (IPGPL), has reportedly said. IPGPL, a consortium of Jawaharlal Nehru Port Trust and Kandla Port Trust, are constructing 2 terminals (5 berths) at Chabahar Port.

India has been fitting machineries to 640-metre long container terminals and

another 600 metre-long multi-purpose made terminals are yet to start functioning, container terminal. It is expected that the trade route between India and Afghanistan was opened recently using other cargo handling capacity would be 85 million ton (mt).

Keeping all these developments in mind, the resurgence in development activity in the Afghan Hajigak region, Swaraj, India's External Affairs Minister especially rail connectivity till the Afghan-Iranian border, be far behind?

Chabahar Port terminals are being constructed according to the specifications agreed upon under the inter-governmental memorandum of understanding (MoU) signed between India and Iran on May 5, 2015. This involves investment of nearly \$85 million for procurement of port equipment. Besides, India has agreed to provide \$150-million credit facility to Iran through the Export-Import Bank of India (EXIM Bank) for Chabahar Port development.

The Indian Ministry of External Affairs (MEA) said in a statement, "The shipment is part of a commitment made by the Government of India to supply 1.1 million tons (mt) of wheat for the people of Afghanistan on grant basis." Six more wheat shipments would be sent to Afghanistan over the next few months. According to the MEA, this would also reaffirm India's commitment to continue their co-operation for the benefit

First, the good tidings for Chabahar. Notwithstanding the fact that the Indian-

and prosperity of the people of Afghanistan the Kerman-Bam-Zahedan railway and its future connection to Chabahar, this port will not fully recognised until a joint Afghan-

For the time being, after unloading connect to the Trans-Iranian Railway. Soviet project, between 1963 and 1965, at Chabahar, the cargo would be carried. Meanwhile, it was reported in the Iranian carried out an extensive study which mapped through the road network all along to the local media that Hossein Ashouri, the deputy and described the deposits in some detail. Afghan destination.

The MEA statement clarified that affairs, has said that the rail has reached the mineral potential of the region, and this was the first shipment that went to the borderline while the Afghan section is estimated the Hajigak resources at 1.8 billion Afghanistan through the Chabahar Port after expected to be completed by the end of 2018 tons of iron ore with a concentration of the Trilateral Agreement on Establishment of Ashouri added that infrastructure has approximately 62 percent Fe. This assessment International Transport and Transit Corridor been completed on the Afghan side though ranks the Hajigak deposits as world class. was signed during Indian Prime Minister pavement lines need to be constructed. The presence of coking coal nearby at Narendra Modi's May 2016 Iran visit.

The rail link covering 191 kilometres Shabashak and the world-class ranking of The MEA statement considered this starts from the Khawaf region in Iran the iron ore resources combine to make the shipment of wheat as a landmark moment and ends in the Afghan city of Herat. Hajigak deposits an exceptionally favourable it would pave the way for operationalisation Afghanistan's section covers 114 km. It is target for economic development.

of the Chabahar Port as an alternate, reliable the first railroad between the 2 neighbouring The multi-billion dollar Hajigak project and robust connectivity for Afghanistan. It countries, whose annual trade hits billions of dollars interlinked with the Chabahar project. A would open up new opportunities for tradedollars annually.

and transit to and from Afghanistan and It is also to be noted that logistics are Steel Authority of India (SAIL), had won enhance trade and commerce between the improving in Afghanistan through initiatives the concession for 3 iron ore mines in the countries and the wider region. from the Asian Development Bank (ADB). Hajigak region in 2011. They were slated to

In the aftermath of Modi's visit, a meeting To improve transport and connectivity, invest \$10.8 billion for setting up steel and of transport ministers of the 3 countries was the ADB said that the bank would continue power plants, besides mining the deposit. held in New Delhi on September 28, 2016 development of the road network, including However, in view of the security risks in the where views were exchanged on the next the Central Asia Regional Economic wake of the withdrawal of US troops, there steps for the implementation of the trilateral Cooperation or CAREC regional corridors, was a go-slow on the project along with a agreement. It was, inter alia, decided to help improve regional trade and local reported scale-down in investments by a convene an expert-level meeting of senior growth. recession-hit industry.

officials of the 3 countries at Chabahar; According to this Manila-based Asian In 2015, Afghan officials claimed that evolve protocols on Transport and Transit, lending institution, support for operation and new mining laws passed by the Afghanistan Ports, Customs Procedures and Consular maintenance would ensure the road network government would make it easier for Indian Affairs; and to organise a connectivity event would provide sustainable benefits. ADB's companies to operate there. Apart from that, at Chabahar, Iran, involving all stakeholders key support to the transport sector would there were issues of evacuation, rail links, in a bid to increase awareness about the new include establishment of the first railway line logistics planning as well as technical issues. opportunities offered by Chabahar Port.

#### Afghan rail initiatives

As far as New Delhi is concerned, one of Transport Sector Master Plan Update would and Europe. This route would pass through the basic economic parameters for Chabahar be developed to cover roads, railways, civil Afghanistan, Turkmenistan, Azerbaijan, Port is to link up with the mineral-rich aviation, urban transport, trade logistics, and Georgia and Turkey, and could be reducing Hajigak region of Afghanistan. Keeping that other related operations in the next 20 years barriers facing transit trade and develop a factor in mind, India has been finalising a Four regional airports in Afghanistan, which Custom Procedure Integration in the region. plan to construct a 900-km railway line that handle more than double the passenger According to the Afghan Chamber of will connect the port of Chabahar, being built volumes seen during pre-upgrade levels, with Commerce and Industry, the road-cum-rail with Indian help, to the mineral-rich Hajigak also be revamped. route would provide an alternative and safe

region of Afghanistan. conduit, bypassing some of Afghanistan's neighbouring countries, which were not an agreement was signed to build the new The Hajigak iron ore deposits are located amenable to trading relations.

Chabahar-Zahedan railway, as part of in the mountainous Bamyan province, 130 Therefore, till the Hajigak region is the North-South Transport Corridor by km west of the Afghanistan capital, Kabul, linked with Chabahar Port via rail, full Indian Railways' public sector unit, Ircon These are one of several iron ore deposits economic benefit cannot not be derived out International. within this area but is the largest located too of the project.

The establishment of Chabahar's date. In the words of Gupta, "the ball has been connection to the country's railway plan is Iron occurrences were observed during set rolling and journey has been initiated via under consideration. With completion of the initial geological mapping of the area in road. How can the rail link be far behind?"

# Global crude steel output dips 1.5% in Sep m-o-m

Sanjoy Bag

World crude steel production for the 66 countries reporting to the World Steel Association (Worldsteel) decreased by 1.50 percent to 141.43 million tons (mt) in September 2017 compared to 143.58 mt reported in August 2017. Crude steel production for September 2017 went up by 6.01 percent compared to 133.41 mt in September 2016. In September 2017, Asia produced 97.93 mt of crude steel, an increase of 5.22 percent over the same month in 2016. The European Union (EU) produced 13.79 mt of crude steel in September 2017, up by 3.96 percent compared to the same month in 2016. North America's crude steel production in September 2017 was at 9.37 mt, an increase of 4.73 percent over the corresponding month of 2016. South America, Africa and Middle East had produced 3.74 mt, 1.15 mt and 0.38 mt during September 2017, an 11.76 percent increase over the same month last year. Pakistan produced 0.46 mt of crude steel in September 2017, a whopping increase of 76.92 percent compared to the same month in 2016. In the EU, Germany produced 3.51 mt of crude steel in September 2017, which is 8.1 percent higher compared to the same month in 2016. Italy produced 2.17 mt of crude steel in September 2017, up by 8.3 percent compared to that in September 2016. Spain's crude steel production was 1.25 mt in September 2017, increase of 6.7 percent compared to September 2016. France produced 1.32 mt of crude steel in September 2017, increase of 3.2 percent compared to that in September 2016. In September 2017, Russia produced 6.03 mt of crude steel, up by 5.8 percent compared to the same month in 2016. Ukraine's production was 1.85 mt in September 2017; down -0.8 percent over the same month in 2016. In September 2017, the US produced 6.72 mt of crude steel, 8.6 percent higher compared to September 2016 and Turkey's crude steel production for September 2017 was 3 mt. The 66 countries' utilisation ratio of crude steel capacity in September 2017 was 73.5 percent and is just 2.8 percent higher of 2016. South America, Africa and Middle East had produced 3.74 mt during September 2017, an 11.76 percent increase over the same month last year.

World crude steel production

in '000 tons

|                     | Sep-16 | Oct-16 | Nov-16 | Dec-16 | Jan-17 | Feb-17 | Mar-17 | Apr-17 | May-17 | Jun-17 | Jul-17 | Aug-17 | Sep 17 / Sep 16<br>(% change) |
|---------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-------------------------------|
| (XURSHDQ 8QLRQ      |        |        |        |        |        |        |        |        |        |        |        |        |                               |
| 2WKHU (XURSH        |        |        |        |        |        |        |        |        |        |        |        |        |                               |
| & , 6               |        |        |        |        |        |        |        |        |        |        |        |        |                               |
| 1RUWK \$PHULFD      |        |        |        |        |        |        |        |        |        |        |        |        |                               |
| South America       |        |        |        |        |        |        |        |        |        |        |        |        |                               |
| Africa              |        |        |        |        |        |        |        |        |        |        |        |        |                               |
| Middle East         |        |        |        |        |        |        |        |        |        |        |        |        |                               |
| \$IULFD 0LGGOH (DVW |        |        |        |        |        |        |        |        |        |        |        |        |                               |
| &KLQD               |        |        |        |        |        |        |        |        |        |        |        |        |                               |
| ,QGLD               |        |        |        |        |        |        |        |        |        |        |        |        |                               |
| -DSDQ               |        |        |        |        |        |        |        |        |        |        |        |        |                               |
| 6RXWK .RUHD         |        |        |        |        |        |        |        |        |        |        |        |        |                               |
| 3DNLVWDQ            |        |        |        |        |        |        |        |        |        |        |        |        |                               |
| 7DLZDQ &KLQD        |        |        |        |        |        |        |        |        |        |        |        |        |                               |
| 7KDLODQG            |        |        |        |        |        |        |        |        |        |        |        |        |                               |
| 9LHWQDP             |        |        |        |        |        |        |        |        |        |        |        |        |                               |
| Asia                |        |        |        |        |        |        |        |        |        |        |        |        |                               |
| Oceania             |        |        |        |        |        |        |        |        |        |        |        |        |                               |
| 5HVW RI WKH ZRUG    |        |        |        |        |        |        |        |        |        |        |        |        |                               |
| H[FHSW &KLQD        |        |        |        |        |        |        |        |        |        |        |        |        |                               |
| World               |        |        |        |        |        |        |        |        |        |        |        |        |                               |

Source: WSA

# Major ports' iron ore handling up 23.5% in Apr-Sept

Steel Insights Bureau



The 12 major Indian ports handled ports showed a rise of 23.5 percent during September, 2017, against 4 mt handled 326.38 million tons (mt) of total April-September, 2017. The major ports during April-September, 2016, traffic during April-September, together handled 21.7 mt of iron ore during Thermal coal handling by the major 2017, about 3.23 percent higher than 316.14 April-September, 2017 compared to 17.57 ports was down 16.35 percent during April-September, 2016, mt handled in the corresponding period of September, 2017. Movement of thermal coal according to data released by the India 2016. through these ports decreased to 41.31 mt recorded during April-September, 2017, compared to 49.39 mt achieved in the corresponding period of the previous fiscal.

Paradip Port handled the highest volume of iron ore, at 5.44 mt, during April- to 49.39 mt achieved in the corresponding period of the previous fiscal.

7UDI&F KDQGOHG DW PDMRU SRUWV GXULQJ \$SULO WR 6TH SEPTEMBER, 2017 vis-a-vis April to September, 2016)

(in ' 000 tons)

| Ports                     | April to 5 GRV G O D G T V T C b y E |      | % variation against RTGX [GCT V T C b y E |
|---------------------------|--------------------------------------|------|-------------------------------------------|
|                           | 2017*                                | 2016 |                                           |
| .2/. \$7\$                |                                      |      |                                           |
| . RONDWD 'RFN 6\ VWHP     |                                      |      |                                           |
| + DOGLD 'RFN & RPSOH[     |                                      |      |                                           |
| 727\$/ .2/. \$7\$         |                                      |      |                                           |
| 3\$5\$ ',3                |                                      |      |                                           |
| 9,6\$. + \$3\$71\$0       |                                      |      |                                           |
| . \$0\$5\$-\$5 (1125(     |                                      |      |                                           |
| & +(11\$,                 |                                      |      |                                           |
| 9 2 & +, '\$0 % \$5\$1\$5 |                                      |      |                                           |
| & 2 & +, 1                |                                      |      |                                           |
| 1(: 0\$1*\$ /25(          |                                      |      |                                           |
| 02508*\$2                 |                                      |      |                                           |
| 080 % \$,                 |                                      |      |                                           |
| -137                      |                                      |      |                                           |
| . \$1' / \$               |                                      |      |                                           |
| 727\$/                    |                                      |      |                                           |

(\*) Tentative

The major ports handled a total of 24.78 mt of coking coal during April-September, 2017, up 3.29 percent compared to 23.99 mt during the same period last year.

Among the major ports, Paradip Port had the distinction of handling the highest volume of thermal coal, at around 12.11 mt, during the first 6 months of the present fiscal. The port had handled 13.92 mt during April-September, 2016.

Movement of container traffic in terms of tonnage rose 6.6 percent to 66 mt during April-September, 2017 compared to 61.91 mt during the corresponding period of 2016.

Altogether, 7 major ports showed positive growth in traffic handling during April-September, 2017, while the remaining 5 showed negative growth on a year-on-year basis.

In terms of growth, Cochin Port topped the list with 19.62 percent increase in cargo throughput, while Chennai's growth was the lowest, at 1.2 percent, during April-September, 2017. In terms of traffic volumes, Kandla Port clinched the top rank with crude oil volumes of 29.96 mt recorded during April-September, 2017. ■

# Railways' Sept iron ore handling up 9% y-o-y

Steel Insights Bureau

Railways transported 11.65 million tons (mt) of iron ore for exports, steel plants and for other domestic use in September 2017, up 9.28 percent from 10.66 mt in September 2016, as per provisional information available with Steel Insights.

Further, revenue earnings of Railways from transportation of iron ore for exports, steel plants and for other domestic use were up 19.32 percent to ₹700.31 crore in September 2017, compared to ₹686.89 crore in the corresponding month of last year.

During April-September, 2017, transportation of iron ore for exports, steel plants and for other domestic use by Railways stood at 70.86 mt, up 12.15 percent from 63.18 mt in April-September, 2016. Meanwhile, revenue earnings of Railways from transportation of iron ore for exports, steel plants and for other domestic use during the first 6 months of financial year 2017-18 stood at ₹4,442.63, up 19.88 percent compared to ₹3,705.64 crore in the corresponding period of 2016-17.

During April-September, 2017, transportation of coal by Railways stood at 258.66 mt, up 0.94 percent from 256.25 mt in April-September, 2016.

Meanwhile, revenue earnings of Railways from transportation of coal during the first 6 months of 2017-18 stood at ₹2,554.98 crore, up 8.09 percent compared to ₹2,086.29 crore in the corresponding period of 2016-17.

Meanwhile, Railways transported 91.43 mt of various commodities in September 2017, up 5.88 percent compared to 86.35 mt in September 2016, while revenue earnings of ₹1,139.51 crore from transportation of these commodities in September 2017 were also up 5.76 percent from ₹7,695.88 crore earned in September 2016.

Cement transported through Railways stood at 8.09 mt in September 2017, down 8.85 percent from 8.16 mt in September 2016. Revenue earnings of Railways from transportation of cement were also down 8.6 percent to ₹596.68 crore in September 2017 from ₹652.83 crore in September 2016.

Commodity-wise revenue earnings of Railways  
(September 2017)

| COMMODITY                     | TONNAGE (in million) |         | EARNINGS (in cr) |         |
|-------------------------------|----------------------|---------|------------------|---------|
|                               | 2016-17              | 2017-18 | 2016-17          | 2017-18 |
| &RDO                          |                      |         |                  |         |
| 7RWDO IRU VWHHO SODQWV        |                      |         |                  |         |
| &RDO IRU ZDVKHULHV            |                      |         |                  |         |
| 7RWDO IRU SRZHU KRXXHV        |                      |         |                  |         |
| 7RWDO IRU SXEOLF XVH          |                      |         |                  |         |
| 7RWDO                         |                      |         |                  |         |
| 3LJ LURQ DQG ILQLVKHG VWHHO   |                      |         |                  |         |
| L IURP VWHHO SODQWV           |                      |         |                  |         |
| LL IURP RWKHU SRLQWV          |                      |         |                  |         |
| 7RWDO                         |                      |         |                  |         |
| ,URQ RUH                      |                      |         |                  |         |
| L IRU H[SRUW                  |                      |         |                  |         |
| LL IRU VWHHO SODQWV           |                      |         |                  |         |
| LLL IRU RWKHU GRPHVWLF XVHUV  |                      |         |                  |         |
| 7RWDO                         |                      |         |                  |         |
| &HPHQW                        |                      |         |                  |         |
| )RRGJUDLQV                    |                      |         |                  |         |
| )HUWLOLJHUV                   |                      |         |                  |         |
| 0LQHDO 2LO 32/                |                      |         |                  |         |
| &RQWDLQHU 6HUYLEF             |                      |         |                  |         |
| L 'RPHVWLF FRQWDLQHUV         |                      |         |                  |         |
| LL (;,0 FRQWDLQHUV            |                      |         |                  |         |
| LLL 7RWDO                     |                      |         |                  |         |
| %DODQFH RWKHU JRRGV           |                      |         |                  |         |
| 7RWDO UHYHQXH HQUQLQJ WUDIILF |                      |         |                  |         |

Commodity-wise revenue earnings of Railways  
(April to September 2017)

| COMMODITY                     | TONNAGE (in million) |         | EARNINGS (in cr) |         |
|-------------------------------|----------------------|---------|------------------|---------|
|                               | 2016-17              | 2017-18 | 2016-17          | 2017-18 |
| &RDO                          |                      |         |                  |         |
| 7RWDO IRU VWHHO SODQWV        |                      |         |                  |         |
| &RDO IRU ZDVKHULHV            |                      |         |                  |         |
| 7RWDO IRU SRZHU KRXXHV        |                      |         |                  |         |
| 7RWDO IRU SXEOLF XVH          |                      |         |                  |         |
| 7RWDO                         |                      |         |                  |         |
| 3LJ LURQ DQG ILQLVKHG VWHHO   |                      |         |                  |         |
| L IURP VWHHO SODQWV           |                      |         |                  |         |
| LL IURP RWKHU SRLQWV          |                      |         |                  |         |
| 7RWDO                         |                      |         |                  |         |
| ,URQ RUH                      |                      |         |                  |         |
| L IRU H[SRUW                  |                      |         |                  |         |
| LL IRU VWHHO SODQWV           |                      |         |                  |         |
| LLL IRU RWKHU GRPHVWLF XVHUV  |                      |         |                  |         |
| 7RWDO                         |                      |         |                  |         |
| &HPHQW                        |                      |         |                  |         |
| )RRGJUDLQV                    |                      |         |                  |         |
| )HUWLOLJHUV                   |                      |         |                  |         |
| 0LQHDO 2LO 32/                |                      |         |                  |         |
| &RQWDLQHU 6HUYLEF             |                      |         |                  |         |
| L 'RPHVWLF FRQWDLQHUV         |                      |         |                  |         |
| LL (;,0 FRQWDLQHUV            |                      |         |                  |         |
| 7RWDO                         |                      |         |                  |         |
| %DODQFH RWKHU JRRGV           |                      |         |                  |         |
| 7RWDO UHYHQXH HQUQLQJ WUDIILF |                      |         |                  |         |

# Tata Steel reports consolidated net profit in Q2



Steel Insights Bureau

**T**ata Steel reported a consolidated net profit of ₹1,017 crore in the September quarter, as against a loss of ₹49.4 crore in the corresponding period last year.

At the consolidated level, this is a profit for 2 consecutive quarters; it reported a net profit of ₹921 crore for the June quarter.

The company that has lined up for buying out 4 debt-laden steel companies, has also said it is committed to the second phase of expansion at its Kalinganagar (Jajpur district) unit in Odisha, with a bullish outlook on demand.

“Our discussion for expansion at Kalinganagar by 5 million tons (mt) is at an advanced stage and we do not see any delay in this. Engineering at the facility has started brownfield expansion from the management and the timeline provided should remain the same,” said T V Narendran, Managing Director for Tata Steel India operations.

Hence, an assurance of brings more clarity on the company’s capacity building strategy.

The blast furnaces at Kalinganagar’s first phase (3 mt) were at 100 percent utilisation and the downstream facilities at 95 percent, the management said.

Tata Steel is among aggressive bidders for some of the large NCLT-listed entities. It is the only company which has made expressions of interest for all 4 assets — Monnet Ispat, Essar Steel, Electrosteel and Bhushan Steel. The company’s consolidated quarterly revenues were at ₹32,464 crore, up 20 percent over a year and 5 percent over the June quarter. Consolidated deliveries were 6.45 mt, up 15 percent year-on-year; India deliveries were 48 percent of the total. “Our view that growth at India operations was broad-based, with all verticals registering strong performance,” said Narendran.

The automotive segment grew 34 percent over a year before, with a focus on new grade development and vehicle models. Sales in the branded products and retail solutions segment grew 14 percent, with strong volume growth in emerging customer accounts. The industrial products, projects and exports segment grew 11 percent, including 60 percent more in targeted value-added and new segment sales. The company said it developed 27 new products across segments in the year.

Consolidated EBITDA (earnings before interest, taxes, depreciation and amortisation) was `4,726 crore, the margins up 14.6 percent. Sequentially, however, this was lower from the `4,939 crore in the June quarter, “due to seasonally weaker performance in our European operations”, said Chatterjee.

For Europe, revenues were higher by 32 percent over a year, reflecting higher deliveries and sales of differentiated products.

EBITDA was £89 million, down £38 million from a year before, primarily due to lower spreads with higher raw material prices, said the management.

“In a relatively stable market environment, we continued to strengthen our sales mix with deliveries of higher value differentiated products increasing by almost 200 basis points in the past year to about 38 percent of total sales,” said Hans Fischer, Managing Director of Tata Steel Europe. This was made possible by new products, of which 5 were launched in the quarter, he said.

Tata Steel India production rises 18% in Apr-Sept

Tata Steel said its saleable steel production in the country rose by 18 percent to 5.98

# Tata Steel elevates Narendran as global CEO & MD

Tata Steel Ltd has promoted T V Narendran as Chief Executive Officer and Managing Director globally.

Narendran, 52, a company veteran who joined the Tata ranks in 1988, was serving as Managing Director (India and South East Asia), for the past 4 years.

Tata Steel executed and commissioned the Kalinganagar steel plant during Narendran's tenure as managing director and also enhanced the plant's ability to serve higher value segments like steel for automobiles, the company said in an exchange filing.

Tata Steel and German major Thyssenkrupp AG had last month announced a preliminary agreement to merge their European steel operations, creating the continent's second-largest steelmaker after ArcelorMittal SA.



million tons (mt) in April-September this year compared to 5.06 mt in the same period last year.

The saleable steel output in the July-September quarter of the current fiscal was 3.02 mt (provisional) against 2.72 mt in the corresponding quarter of the previous fiscal, Tata Steel said in a statement.

The company's sales in India were at 5.88 mt in April-September this fiscal against 4.77 mt in the first half of FY2017.

“Q2 FY18 sales registered a volume growth of 12 per cent (q-o-q) and 17 per cent (y-o-y),” the statement said.

The Tata Steel Group, with an annual crude steel capacity of 27.5 million tons per annum as on March 31, 2017 has operations in 26 countries.

Tata Steel strengthens position in low-carbon future tech

Meanwhile, Tata Steel has said it is strengthening its position in low-carbon future technology, which has potential to reduce energy use and carbon emissions by at least 20 percent, as well as reducing steel-making costs.

Tata Steel is at the forefront of lower-carbon steel innovation by acquiring the full intellectual property rights in a revolutionary technology.

The company has been testing the ground breaking technology called HIsarna at its IJmuiden steelworks in the Netherlands. HIsarna has the potential to reduce energy use and carbon emissions by at least 20 percent, as well as reducing steel making costs through lower-priced raw materials, up to half of which could be recycled scrap steel, the company said in an official statement.

HIsarna is a new technology in the steel making process which combines Tata Steels cyclone converter furnace with Rio Tinto's smelter. Tata Steel has already acquired the Rio Tinto's smelter technology and intellectual property rights required to operate the HIsarna process.

“The acquisition of full intellectual property rights for HIsarna, is a significant step towards establishing our pioneering credentials in the steel manufacturing space,” Tata Steel managing director T V Narendran said. ■

| Heading                |                     |           |          |               |             |               |
|------------------------|---------------------|-----------|----------|---------------|-------------|---------------|
|                        | Items               | Quarterly |          |               | Half-yearly |               |
|                        |                     | Q2 FY17   | Q1 FY18  | Q2 FY18       | H1 FY17     | H1 FY18       |
|                        |                     | (Actual)  | (Actual) | (Provisional) | (Actual)    | (Provisional) |
| , Q G L D              | 3 U R G X F W L R Q |           |          |               |             |               |
|                        | Sales               |           |          |               |             |               |
| ( X U R S H            | 3 U R G X F W L R Q |           |          |               |             |               |
|                        | Sales               |           |          |               |             |               |
| 6 R X W K ( D V W Asia | 3 U R G X F W L R Q |           |          |               |             |               |
|                        | Sales               |           |          |               |             |               |

# JSW consolidated Q2 PAT up 29% to `836 cr

Steel Insights Bureau

JSW Steel reported a consolidated profit after tax (PAT) of `836 crore in the second quarter ending September 30, 2017, of the current fiscal (2017-18), showing a 29 percent jump against `647 crore recorded in the same quarter of the previous fiscal, after incorporating the financials of subsidiaries, joint ventures and associates, per a company statement.

The company's consolidated revenues from operations increased by 17 percent y-o-y to `16,818 crore (`14,421 crore) for the quarter under review. Operating EBITDA increased by 4 percent y-o-y to `3,036 crore, implying a margin of 18.1 percent. Operating EBITDA increased by 16 percent q-o-q, while PAT increased by 34 percent q-o-q.

On a standalone basis, the company reported a 26 percent increase in PAT in Q2 y-o-y to `845 crore (671 crore). Operating EBITDA increased by 33 percent q-o-q, while PAT increased by 102 percent q-o-q.

Standalone revenues from operations for the quarter stood at `14,956 crore, up 12 percent y-o-y. Despite higher prices of fluxes and power cost, operating EBITDA for the quarter increased by 9 percent y-o-y to `2,927 crore, aided by higher sales volumes and lower prices of key inputs like coking coal, with EBITDA margin of 19.6 percent.

Crude steel production during the quarter stood at 3.94 million tons (mt). Production was impacted by continued water shortage in the south, particularly in Tamil Nadu, as well as availability of iron ore at Vijayanagar.

Saleable steel sales volume for the quarter was 3.92 mt, up by 2 percent y-o-y. The company liquidated a large part of the inventory that was built up during the previous quarter in the run-up to GST implementation and, consequently, sales

volumes increased 12 percent q-o-q during the quarter.

Aided by an increase in domestic demand and favourable international markets during the early part of the quarter, the company achieved its highest-ever consolidated quarterly saleable steel sales of 3.96 mt. As the impact of GST rollout on channel sales began to stabilise, JSW Steel's consolidated domestic sales volume increased by 10 percent q-o-q, while retail sales have grown by 24 percent on a q-o-q basis. With better international demand, the company focused on increasing exports volumes which surged by 33 percent q-o-q. The company remained focused on enriching the product mix and the overall sales of value-added and special products (VASP) and special products grew by 17 percent q-o-q.

JSW Steel Coated Products: During the quarter, JSW Steel Coated Products registered a production volume (galvanised products) of 0.43 million tons and sales volume of 0.56 mt. Revenues from operations and operating EBITDA for the quarter stood at `3,416 crores and `141 crore respectively. Net profit after tax was `53 crore for the quarter.

US Plate and Pipe Mill: The US-based Plate and Pipe Mill facility produced 52,315 net tons of plates and 11,488 net tons of pipes, reporting a capacity utilisation of 22 percent and 8 percent, respectively during the quarter. Sales volumes for the quarter stood at 49,226 net tons of plates and 11,822 net tons of pipes. It reported an EBITDA of \$1.0 million for the quarter.

Global outlook

The global growth outlook is broad-based and remains positive over the near term as economic indicators across regions point towards an accelerating momentum, the company said in a statement. Growth in the US remains firmly on course with strong

business and consumer confidence, supported by accommodative financial conditions. Euro area growth continues to recover, aided by steady domestic demand and higher exports amidst expansionary monetary policy, and lower political risk and policy uncertainty. Outlook for Japan is also improving with strengthening global demand and supportive fiscal stance, the company said, adding that economic growth in China has been strong with high public investments and the momentum is expected to continue with robust domestic demand.

The World Steel Association (WSA) expects world steel demand to grow at 2.8 percent y-o-y in 2017 – more than double the growth rate expected at the beginning of the year. Steel exports from China have reduced further in the last quarter with closure of inefficient production facilities and continued strong domestic demand. Exports from Japan reduced slightly with stronger domestic demand, but Korean exports continue to remain at elevated levels with weakness in the domestic market. Steel prices have strengthened with improved demand outlook, production discipline and support from raw material prices.

Steel imports into India have increased sharply in the last quarter displacing domestic volumes. The Q2FY18 imports run rate was ~12mt (up by 59 percent q-o-q). A surge in imports of flat products in Q2 met almost the entire q-o-q growth in flat products consumption in the quarter. Rising imports of defective materials and imports of substandard products under the garb of prime products, especially in the coated segment, is alarming, the company warned.

Crude steel production in India increased by 4.5 percent y-o-y in H1FY18 and the apparent finished steel consumption grew by 4.3 percent y-o-y. However, consumption of domestically produced steel increased only by 2.8 percent y-o-y as imports surged rapidly.

Steel demand is expected to improve further with the government focus on infrastructure build-up like roads, Dedicated Freight Corridors (DFCs), water and gas pipelines, metro, irrigation projects, solar energy, power T&D etc. Bank recapitalisation will pave the way for an uptick in credit growth and possible restart of the investment cycle, feel company officials.

# Tata Metaliks net zooms

Steel Insights Bureau

Pig iron and ductile iron water pipe maker Tata Metaliks Limited (TML) has reported a 54 percent year-on-year increase in quarterly net profit, helped by higher sales and lower input cost.

For the quarter ended September 30, net profit increased to ₹33.5 crore from ₹21.73 crore a year ago, the company said. Its revenue stood at ₹450 crore compared with ₹342.85 crore in the same quarter of FY17, as pig iron sales jumped 30 percent on a sequential basis over the first quarter of the current fiscal, mainly on account of firming up of prices in August and September.

TML also reaped the benefit of improved cost efficiencies and higher availability of pig iron due to modernisation of one of its blast furnaces.

The company's half-yearly net profit rose 13.17 percent to ₹64.1 crore compared with ₹56.7 crore a year ago, while revenue for the period touched ₹858.25 crore compared with ₹677.20 crore in H1 of 2016-17.

Drop in sales of DI Pipe in the current quarter compared to previous quarter was due to two main reasons: 1) lack of clarity on tax impact post GST implementation (most of the DI pipes for projects were earlier exempt from excise duty) and 2) Monsoons - work at project sites are usually impacted because of

rain, the company said in a statement. Revenue from pig iron sales in Q2 FY'18 was 30 percent higher than Q1 FY'18 as a result of rising pig iron prices in August and September on the back of higher international coke prices, as also due to improved availability of Pig Iron volumes due to modernization of one of the blast furnaces, the company said in a statement.

EBITDA increase in Q2 is mainly on account of much higher sales of pig iron as also due to improved spread between coke price and Pig Iron prices, coupled with improved cost efficiencies.

Coal and coke prices have started softening in Q3 which should help the company reduce its costs but there could also be pressure on the pig iron market. Also, the benefits from 10 MW captive power plant which is operating at 100 percent capacity is expected to mitigate the impact of falling commodity prices.

Sandeep Kumar, Managing Director of Tata Metaliks Limited said: "The Company's focus on achieving cost leadership through structural cost reduction projects are helping it deal with market hiccups arising out of volatility in commodity prices and uncertain events like GST. The outlook for H2 looks positive with expected demand pick-up after the monsoons and major festivals but any fall in pig iron prices could be a dampener."



IMFA Q2 net decreases by 82% Q-o-Q Indian Metals & Ferro Alloys Ltd (IMFA), a leading domestic producer of ferro chrome, posted 82 percent slide in net profit to ₹37 crore in the quarter ended September 30, 2017 (Q2 FY18) against ₹42.31 crore in Q2 FY17.

The company revenues went up 7.5 percent to ₹416.03 crore in Q2FY18 against ₹387.58 crore in the previous corresponding period.

IMFA recorded exports of ₹344.94 crore during the quarter under review, which was higher than ₹323.70 crore worth of exports in same period last year.

IMFA's net profit was at ₹107.29 crore in H1FY18 on revenues of ₹842.34 crore. Compared to this, it had posted a net of ₹11.96 crore in H1FY17 on revenues of ₹640.87 crore. The company's exports touched ₹721.78 crore in H1FY18 compared to ₹541.12 crore in H1FY17.

In terms of operation, IMFA reported a ferro chrome production of 58,413 ton, in Q2FY18 down 4 percent from 60,836 ton in Q2FY17. Sales were up 5 percent to 65,013 tons, from 61,762 ton in Q2FY17.

Subhrakant Panda, managing director of IMFA said: "Ferro Chrome has been quite volatile and the Q2 FY18 benchmark price was negotiated at a time when sentiments we were unable to benefit having already finalised the quarterly contracts."

He added that riding on positive sentiments, the pricing of long term contracts in the ongoing quarter is substantially higher and will yield healthy margins. The company also said it expects prices to move within a relatively narrow band, going ahead.

Tata Metaliks H1 performance highlights

| \$OO 2JXUHFV RQH V XQOHVV VQFH18H |          |          |                       |          |          |
|-----------------------------------|----------|----------|-----------------------|----------|----------|
| FY'17                             | H1 FY'18 | H1 FY'17 | Q2 FY'18              | Q1 FY'18 | Q2 FY'17 |
|                                   |          |          | 3LJ LURQ VDOHV W      |          |          |
|                                   |          |          | ', SLSH VDOHV W       |          |          |
|                                   |          |          | ( %, ' 7\$            |          |          |
|                                   |          |          | 'HSUHFLDWLRQ          |          |          |
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|                                   |          |          | ([FHSWLRQDO LWHPV     |          |          |
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# RINL to produce 1 mt of special steel

Steel Insights Bureau

The Visakhapatnam steel plant - Rashtriya Ispat Nigam Limited is planning to produce one million tons of special steel during 2017-18, according to Chairman and Managing Director P. Madhusudan.

"RINL is focusing more on producing Special Steel/Value Added Steel with the introduction of new technologies in the Mills and accordingly strategies were drawn up to improve the RINL market share in the Special Steel segment in the country," he said.

He was speaking in the plant after inaugurating the all-India special steel customers' meet organised by the marketing department of the RINL.

Addressing the customers, Madhusudan said that "Special Steel" plays a significant role in the present market conditions in improving the brand image and would contribute in enhancing the bottom-line of the organization. RINL is presently emphasizing on improving the efficiency and ramping up of production from the expansion units to achieve cost competitiveness, he said.

Madhusudan informed that RINL has completed the modernization of existing units with the blowing-in of Blast Furnace-2 - KRISHNA and expressed confidence that RINL would fare better during the current financial year with the operation of all the three Blast Furnaces. He exhorted the customers to come out with useful deliberations/suggestions to improve the Special Steel products being rolled out from VSP.

He said that the RINL had rolled out spring steel and spring steel flats for the first time in 80/16 size. The plant had also started production of low-alloy steel from the structural and special bar mill. Special steel products would largely cater to the needs of automobile industry, the Railways and engineering industry, he added.

He said special steel would play a

significant role in improving the brand image and bottom line of the RINL. He said the RINL had completed the modernisation of the existing units with the blowing-in of blast furnace two and expressed confidence that the company would fare better during the current year with the operation of all the three blast furnaces.

P. Raychaudhury, the Director (Commercial), and P.K Rath, the Director (Operations) also spoke on the occasion. Special steel customers from automobile industry, Indian Railways, General Engineering etc from all over India took part in the meeting.

Company set to take up coastal shipping of products

RINL - has made all arrangements to undertake coastal shipping of its products to different parts of the country through coastal shipping.

Union Steel Minister Chaudhary Birender Singh flagged off the first consignment of RINL products to various destinations by coastal shipping from Visakhapatnam port. The steel plant has awarded the 25-crore annual contract for dispatching 90,000 tons to Ahmedabad, 75,000 tons to Mumbai and 60,000 tons to Kochi.

The annual contract will be executed by a consortium of Shreya Shipping Logistics Ltd, part of Dubai-based Transworld Group, and Srivalli Shipping, according to RINL officials. Till now, RINL is transporting TMT bars, wire rods and other products to various stockyards by road and rail. Transportation by coastal shipping is expected to reduce logistics cost substantially.

Told to play vital role as a steel producer

Ministry of steel's joint secretary Ruchika Chaudhry has advised RINL to cash in on the huge infrastructure funding by the government to build the infrastructure projects like the national highways, railways

## Special Steel from RINL

- y RINL is planning to produce 1 million tons of Special Steel during 2017-18.
- y RINL has already rolled out Spring Steel and Spring Steel Flats for the first time in 80X16 size.
- y RINL has started producing Low Alloy Steel from Structural and Special Bar Mill.
- y High Carbon Wire Rod sales increased to around 15,000 tons/per month and RINL started producing Low carbon Cold Heading Quality (CHG) grade from Wire Rod Mill-2.
- y The Special Steel products of RINL would largely cater to the needs of Automobile Industry, Indian Railways, General Engineering etc.

and housing for increasing the consumption of steel in the country.

She said that RINL should penetrate Steel into the market by establishing its brand image prominently across the country and should play a vital role as a steel producer. She lauded the steel enterprise for its greenery, environmental protection measures and green initiatives in energy consumption.

Chaudhry said that employees should be given the skill enhancement training to increase operational efficiencies.

Rath assumes charge as Director (Ops)

Pradosh Kumar Rath assumed charge as Director (Operations) at RINL on September 10, 2017. Prior to this Rath was the Executive Director (Works) I/c of RINL.

Rath started his career in RINL-VSP as a Management Trainee in 1983 and has worked in various capacities. He has rich experience in steel making and continuous casting processes. He contributed significantly to the stabilizing and ramping up of production from the new expansion units. He is presently the Chairman of Indian Institute of Metals (IIM), Visakhapatnam Chapter.

# Vedanta PAT up 41% in Q2



Vedanta limited Q2 performance

| Q2                                                                 | 2017-18 | 2016-17 | % change |
|--------------------------------------------------------------------|---------|---------|----------|
| \$ W W U L E X W D E O H S U R I L W<br>D I W H U W D [ H V 3 \$ 7 |         |         |          |
| 1 H W V D O H V<br>7 X U Q R Y H U                                 |         |         |          |
| ( % , 7 ' \$                                                       |         |         |          |

are at advanced stages of being awarded. We expect H2 of this fiscal year to be more robust with the continuing production ramp-up and we continue to maintain a strong balance sheet and remain focused on creating long-term shareholder value,” said Kuldip Karua, CEO, Vedanta Limited.

H1 production of iron ore at 1.9mt

Vedanta Limited produced 1.9 million tons (mt) of iron ore at Karnataka in H1 and expects to achieve full allocation production during Q3. Production from Goa was at 2.6 mt and sales at 2 mt, compared to 2.9 mt and 2.4 mt in H1 of the previous fiscal, respectively. Production and sales at Goa were impacted by the low pricing environment.

The company's production and sales, however, from Karnataka moved up (y-o-y) to 1.9 million tons and 1 million tons.

During the first six months of this fiscal, average oil and gas gross production, including from offshore assets such as Ravva and Cambay, was at 184,062 bpd (barrels of oil equivalent per day), Vedanta said.

Copper production stood at 1, 97,000 tons of cathodes, similar to the first half of the previous fiscal.

Aluminium production increased to 7,53,000 tons in H1, 2017-18 by 39 percent higher (y-o-y), mainly on account of ramp-up of additional pots at Jharsuguda II and BALCO II.

Alumina production was 572,000 tons during the period. Power sales decreased to 4,787 million units mainly due to a temporary coal shortage during the six months under review, it added.

“During the quarter, our zinc, copper and aluminium businesses have delivered a strong production performance. We have also commenced our growth journey on both the exploration and development front in our oil and gas business. We are continuing to realise operational efficiencies across our diversified portfolio and to benefit from a supportive market environment,” said Kaur.

Steel Insights Bureau

Vedanta Limited recorded a 41 percent y-o-y increase in consolidated profit after tax (PAT) at 2,036 crore for the quarter ended September 30, 2017 against 1,525 crore seen in the corresponding quarter last year, the company said in a BSE filing.

Revenues rose 37.37 percent to 21,520 crore in the quarter under review against 15,665 crore seen in the same quarter of the previous fiscal (2016-17).

The EBITDA jumped 24 percent y-o-y to 5,776 crore during the quarter under review against 4,647 crore recorded in the same period last year.

However, other income of the company slipped 36 percent y-o-y to 876 crore.

Revenues in Q2 increased by 37 percent y-o-y on account of higher volume at Copper India, Zinc India, Zinc International, ramp-up at the company's aluminium business and higher commodity prices partially offset by currency appreciation and lower volumes at oil and gas. The net sales of the company stood at 21,520 crore in the second quarter of the current fiscal.

“PAT for the quarter was up 41 percent and EBITDA was up 24 percent compared to last year on the back of solid operational performance in our zinc and copper businesses, supported by strong commodity prices. Key contracts on our announced oil and gas projects

Iron ore sales, production in Q2 and H1

| Particulars (in million dry metric tons, or as stated) | Q2     |        |                | Q1     |                | H1     |        |                |
|--------------------------------------------------------|--------|--------|----------------|--------|----------------|--------|--------|----------------|
|                                                        | FY2018 | FY2017 | % change y-o-y | FY2018 | % change q-o-q | FY2018 | FY2017 | % change y-o-y |
| , 5 2 1 2 5 (                                          |        |        |                |        |                |        |        |                |
| 6 \$ / ( 6                                             |        |        |                |        |                |        |        |                |
| Goa                                                    |        |        |                |        |                |        |        |                |
| . D U Q D W D N D                                      |        |        |                |        |                |        |        |                |
| 3 U R G X F W L R Q R I                                |        |        |                |        |                |        |        |                |
| Saleable Ore                                           |        |        |                |        |                |        |        |                |
| Goa                                                    |        |        |                |        |                |        |        |                |
| . D U Q D W D N D                                      |        |        |                |        |                |        |        |                |
| 3 U R G X F W L R Q                                    |        |        |                |        |                |        |        |                |
| 3 L J , U R Q                                          |        |        |                |        |                |        |        |                |

Expressway built with SAIL steel lands Air Force aircraft

Steel Insights Bureau

| Ports & Terminals          | Q2 FY17<br>(mn tonnes) | Q2 FY18<br>(mn tonnes) | Growth | H1 FY17<br>(mn tonnes) | H1 FY18<br>(mn tonnes) | Growth |
|----------------------------|------------------------|------------------------|--------|------------------------|------------------------|--------|
| +D]LUD GU\ EXON XQLW FDUJR |                        |                        |        |                        |                        |        |
| 9L]DJ GU\ EXON FDUJR       |                        |                        |        |                        |                        |        |
| 3DUDGLS GU\ EXON FDUJR     |                        |                        |        |                        |                        |        |
| &RQVROLGDWHG               |                        |                        |        |                        |                        |        |

Steel Authority of India (SAIL) supplied 83,500 tons of steel for the construction of the Agra - Lucknow expressway which facilitated the touch-and-go landing of planes of the Indian Air Force. SAIL had primarily supplied long products consisting of earthquake resistant TMT bars for this project. SAIL's branded earthquake resistant EQR TMT Bars have superior qualities and come in various customized specifications, the company said in a release.

IAF conducted the touch and go landings on a stretch of the expressway, starting with a 35,000-kg C-130J Super Hercules aircraft. This is the first time that the C-130J transport aircraft joined the Indian Air Force's special drill to land planes on the expressway. The IAF had earlier landed its Mirage 2000 and Sukhoi-30MKI fighter jets on the Yamuna Expressway and the Lucknow-Agra Expressway.

Keeping with the spirit of SAIL's tagline, 'There's a little bit of SAIL in everybody's life', the Company is increasingly focusing to produce world-class steel products befitting the requirements of every segment and market including the specific needs of rural and urban areas to meeting the necessities of niche segments like defense sector too.

SAIL steel has been integral in projects of national importance spanning from Sardar Sarovar Dam, Dhola-Sadiya bridge, Chenani-Nashri tunnel to various projects, power plants and upcoming Statue of Unity at Gujarat.

SAIL supplies special grade steel for INS Kiltan

Steel Authority of India Ltd. (SAIL) has supplied defence grade DMR 249A steel plates for the indigenously built Anti-Submarine Warfare (ASW) stealth corvette

INS – Kiltan which was commissioned on October 16, 2017 into the Indian Navy. SAIL's integrated functioning across all its plants has been able to successfully supply the required quantity of steel for this significant project. DMR 249A is a low carbon micro-alloyed grade of steel with stringent toughness requirement at sub-zero temperature. SAIL developed this warship grade steel plates for the Indian Navy in collaboration with Defence Metallurgical Research Laboratory, Hyderabad.

SAIL has been supplying steel for defence sector for a long time and its steel has been used in various other prestigious ship building projects. SAIL has supplied more than 50,000 metric tons of DMR 249A defence grade steel for naval ships against various requirements.

Essar Ports cargo handling grows 19%

Essar Ports handled 17.35 million tons (mt) of cargo in the first half (H1) of financial year 2017-18, registering a 19 percent growth over cargo handled in the corresponding period of 2016-17. The company also registered a 7 percent growth in cargo handling in the quarter ending September 30, 2017, when compared to the same period in the previous fiscal.

Essar Ports, which recently completed the sale of its Vadinar Oil terminal to Essar Oil Limited as part of Essar Oil's sale to Rosneft, and a consortium led by Trafigura and UCP, has 3 operational ports in Hazira, Vizag and Paradip. All 3 ports registered healthy growth in cargo handling in the first half of the current fiscal. The substantial growth in traffic came on the back of an improving steel market, leading to enhanced capacity utilisation of Essar Steel, which is Essar Port's anchor customer.

The traffic growth was also helped by the 50 percent increase in third-party cargo volumes during the first half of 2017-18. The iron ore terminal at Vizag Port surpassed past benchmarks and was able to deliver productivity in excess of 60,000 tons per day (tpd). The productivity is expected to be in excess of 100,000 tpd in Q3 of 2017-18. In the third quarter, Essar Ports is also expected to commission its terminal facility in Vizag Port on the east coast of India.

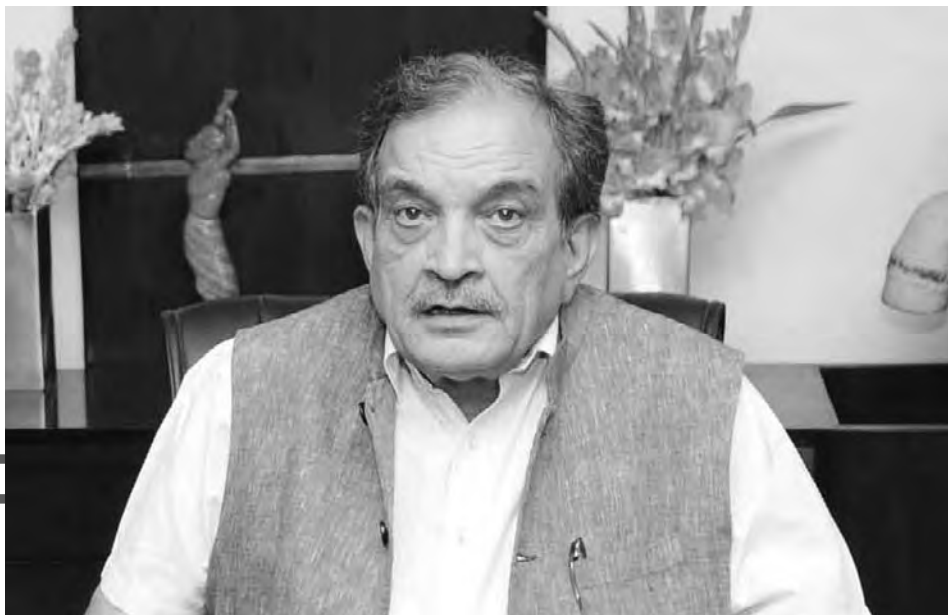
Rajiv Agarwal, CEO & MD, Essar Ports, said: "The first half of FY18, as well as the second quarter of the current financial year, have been very promising for Essar Ports. We completed monetisation of the Vadinar Oil Terminal (VOTL), and our facilities (which now exclude VOTL) handled 17.35 mt of cargo, registering 19 percent growth. We are confident of further growth and performance improvements in the next 2 quarters of the fiscal, as we prepare to commission our project at Vizag."

Tata Steel Kalinganagar unveils largest CDQ unit

In its continuous journey towards improving productivity, energy efficiency and environmental performance, Tata Steel has established India's largest coke dry quenching (CDQ) facility capable of handling 200 tons per hour, at its state-of-the-art greenfield steel plant located at the Jajpur district of Odisha.

CDQ is a heat recovery system to cool the hot coke from coke ovens. It is one of the most renowned energy-efficient and environment-friendly facility in steel production where hot coke removed from coke ovens at a temperature of approximately 1,000°C is cooled and kept dry with inert gas and the resulting steam produced in a waste heat recovery boiler is used to generate electricity.

# India needs to develop 20-25% of steel capacity along coast: Minister



Steel Insights Bureau

Steel minister Birender Singh has said India could aspire to develop 20-25 percent of its steel capacity along the country's coastline by 2025 to meet its export targets.

The minister's comments came during an event when he flagged off the maiden coastal shipment of Rashtriya Ispat Nigam Limited (RINL), the corporate entity of Vizag Steel, marking the steel major's foray into sea trade for its domestic needs.

Speaking on the occasion, the Steel Minister said the Sagarmala Project would transform the logistic sector and change the lives of those living along the country's 7500 km coastline. India could aspire for 25-30 percent of steel capacity to be coastal by 2025 to meet the requirement of steel exports, the minister said. At present, logistic cost in India is amongst the highest in the world but Sagarmala programme has the potential to unlock full potential of India's coast line and waterways and make logistics sector competitive with the world standards, he added.

Coastal shipping is cheaper than road or rail by 60-80 percent and reduces the burden on rail and road transport. He observed that an overall cost saving of around 40,000 crore per annum is estimated from this project by 2025.

While commending RINL for foraying into sea trade to strengthen its relationship with the coastal transportation for domestic

requirements, he said and added that RINL must take its joint ventures at international level to logical conclusion at the earliest. The company must leverage its position in the industry, instead of not being able to fulfill its commitments and agreements," Birender Singh said in an official statement. efficient and safer.

The Steel Minister flagged off maiden management to submit quarterly plans and Coastal Shipping of RINL-Vizag Steel products at Visakhapatnam Port Trust. three months. The Steel Minister will also

The project will be executed by a private consortium to haul 2.25 lakh tonsto get direct feedback about constraints, of VSP products to Ahmedabad, Kochi andbottlenecks and issues of concern.

Mumbai. The RINL is foraying into sea trade for the first time under the Sagarmala project initiated by the Central government. by adopting new technology and adding

The Minister also commissioned the value added products to its basket, like non newly built Turbo Blower-5 on the Thermal corrosive steel products for construction in Power Plant premises. This is a standbycoastal areas.

Turbo Blower to ensure uninterrupted supply SAIL should also focus on product differentiation to create value for the shareholders, thereby creating an exclusive

of Cold Blast to Blast Furnace-3. space for the company. Efficiency improvement and product quality

The Steel Minister has asked Steel Authority of India Limited (SAIL) to increase be focused upon to meet the customer requirements and to improve the financial performance of the company, the steel minister said.

"SAIL must ramp up production from New Rail Mill in Bhilai without wasting any

more time, and meet all the requirements of Steel ministry proposes to set up scrap-based steel plants

its major customers. Company should chalkThe government has announced the proposal



to set up scrap-based steel plants in the northern and western part of the country. The steel ministry proposes to set up steel plants with scrap as the raw material in various parts of north and west India," Chaudhary Birender Singh said in New Delhi.

The minister was addressing the India Segment of 2017 World Recycling Convention, organised by the Bureau of International Recycling (BIR) in partnership with FICCI, the industry body said in a statement. The government expects that in the coming years, 44 percent of the total scrap available in India would be generated at different locations in Jammu and Kashmir, Punjab, Haryana and Delhi would be used to produce steel, the statement said.

In addition, 67 percent of scrap reaches western shores which could be utilized as feedstock to produce steel, the statement added. Singh said that initiative of the government to recycle waste products for the productive purposes would result in saving 65 percent of iron ore.

The government, he said, is adopting a 360 percent holistic approach wherein the recycling industry can assist to achieve the production target by providing raw material for the steel sector. The minister also expressed hopes that legislation would be enacted to ban use of diesel vehicles in India. "... the pending

legislation on the ban on use of 15-year old diesel vehicles in India would be taken up and passed in the winter session of Parliament," the minister said. According to the estimates of the steel ministry, after the ban, the scrap market would become attractive as it would fetch 10 to 15 percent more price for metal scrap.

There is huge potential for e-rickshaws, e-tempos and e-cars in India, he told the PSU and said MOIL must be prepared for harnessing. In fact, the company should innovate to find new uses of manganese ore, apart from steel making, dry cell batteries and paints, he added. Further, he advised MOIL that R&D efforts should aim for development of new technologies rather than updating or buying technologies from others and there is a need to work on utilizing every grade of ore gainfully including the lowest quality ore. In view of growing future prospects for the electric vehicles in the country, the Centre has asked MOIL to increase productivity by upgrading technologies, capacity expansion and acquiring stake in the mid-year review meeting of MOIL, the Minister also asked the PSU to plan for rising requirements from Defence and the building capacities in tungsten mining and electric vehicles market, according to a government release.

In view of the increase in requirement for manganese for steel production capacities the Ministry of Steel said in a statement. "MOIL must make significant contribution to Make in India by developing technology to meet manganese requirement also explore the possibility of acquiring or taking stake in manganese ore assets abroad the minister said.

MOIL must make significant contribution to Make in India by developing technology to meet manganese requirement also explore the possibility of acquiring or taking stake in manganese ore assets abroad the minister said.

# Star Wire (India) orders billet/bloom caster from Primetals

Steel Insights Bureau

Indian steel producer Star Wire Ltd. has placed an order with Primetals Technologies to supply a continuous billet/bloom caster for its steel plant in Ballabgarh, Haryana State.

The caster is designed for the production of 30,000 to 60,000 tons per year, depending on the steel grades cast. It complements the existing ingot casting route. Goals of the project are to further improve the quality of cast products as well as the yield of the production process from liquid steel to ready-to-roll products. The casting process will be fully automated. This is the first continuous billet and bloom caster for an extended steel grade mix worldwide. It also features the first application of dynamic soft reduction for long products in India. The billet/bloom caster is planned to be started up in late 2018.

The new continuous billet/bloom caster will have a machine radius of 12 meters and a metallurgical length of 20 meters. It will produce square billets with dimensions of 145x145 and 195x195 millimeters, with provisions being made for the future production of larger cross sections up to 350x350 millimeters. It will cast a wide variety of steel grades, including austenitic and martensitic heat resistant steels, tool and die steels, ferritic, martensitic and austenitic stainless steels, and other special steels.

Primetals Technologies will be responsible for the basic, detail and process engineering, the complete supply of caster process equipment including technological structure, hydraulic system, subsystems like auto-torch cutting machines and similar equipment. Among the key features are level 2 process automation and technological

packages, like DiaMold mold, the DynaFlex mold oscillator, LevCon automatic mold-level control and DynaGap soft reduction Technologies has commenced operations at the production site of Turkish steel producer Tosyali-Toyo Celik A.Ş. in the city of Osmaniye.

The cold rolling complex consists of a coupled pickling line-tandem cold mill (PL-TCM), a tin-plated steel-sheet continuous annealing line (Tin-CAL) and a temper double-cold-reduction (Temper/DCR) mill. It is designed to produce cold rolled tin-plated, hot-dip galvanized and painted sheets, mainly for export markets as well as for the domestic market. This marks the expansion of Tosyali into the flat product sector. The PL-TCM features the first application of the patented iBox pickling process in the Europe/MENA region. Primetals Technologies had received the order in late 2014.

stainless castings weighing up to 80 tons, Abinsk Electric Steel Works starts up

minimill modernized by Primetals

The minimill modernized by Primetals Technologies was started up at Russian long product producer Abinsk Electric Steel. The modernization included the company's electric arc furnace, ladle furnace and 6-strand continuous billet caster. The aim of the project was to increase the production capacity of billets with cross sections of 130 and 150 millimeters from 1.2 to 1.5 million metric tons per year. Additionally, Abinsk is now able to produce more quality steel grades, such as high carbon steel for wire and spring steel. Moreover, the new technology not only increases the availability of the slab works, but also decreases maintenance costs. Conversion costs also are substantially lowered. The energy requirement of the electric arc furnace is reduced from 410 to 370 kilowatt hours per metric ton.

with the purpose of further improving operational efficiency and product quality.

Karl Purkarthofer, Senior Vice President and Global Head of Metallurgical Services at Primetals Technologies stated: "This cooperation is a major milestone for technology based maintenance services in China and will elevate the successful partnership between Tangsteel and Primetals Technologies to a new level."

Tosyali-Toyo starts sheet production with cold rolling

A cold rolling complex supplied by Primetals Technologies has commenced operations at the production site of Turkish steel producer Tosyali-Toyo Celik A.Ş. in the city of Osmaniye.

The cold rolling complex consists of a coupled pickling line-tandem cold mill (PL-TCM), a tin-plated steel-sheet continuous annealing line (Tin-CAL) and a temper double-cold-reduction (Temper/DCR) mill. It is designed to produce cold rolled tin-plated, hot-dip galvanized and painted sheets, mainly for export markets as well as for the domestic market. This marks the expansion of Tosyali into the flat product sector. The PL-TCM features the first application of the patented iBox pickling process in the Europe/MENA region. Primetals Technologies had received the order in late 2014.

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# Waste-to-wealth is Hind Copper's new biz mantra

**H**industan Copper Limited (HCL), a public sector enterprise and India's only vertically integrated copper producing company, with facilities that encompass mining, beneficiation, smelting, refining and casting of refined copper into downstream products, is looking at a major expansion by 2020. This includes mine expansion, enhancement of capacity at the copper major's various plants, ranging from beneficiation to refining and, most importantly, extracting value out of the waste generated from the copper ore mined. Consequently, future plans include production of newer metals and minerals like nickel, gold and silver as well as glass-grade silica from copper ore tails, (a waste generated during the copper ore beneficiation process) and obtaining value out of waste rock.

Santosh Sharma, the newly-appointed Chairman-cum-Managing Director and architect of the new business ventures, tells Madhumita Mookerji, the new business areas are likely to generate additional revenues of `55-60 crore per annum. Excerpts from an interview:

Tell us a bit about Hindustan Copper's expansion plans

We have taken up many projects currently which include increasing our mining, beneficiation, smelting and refining capacities.

Where mining is concerned, HCL's

current ore production capacity is 3 mt per annum but we have plans to increase this to 12 mt by 2022.

We have 2 mines in the Khetri mine belt where we plan expand to 3.5 mt from the present 1.1 mt.

Another volume of 3.5 mt will come

from the Jharkhand belt. Here, we have mining-cum-beneficiation and processing facilities. However, we are not directly operating it at present but have outsourced it to a mine developer. In the Jharkhand mines area, we have the Surda mine and adjoining it are Kendadih and Rakha. Alongside Rakha, there are Chhapri and Siddheswari. The total capacity of this entire belt, post-expansion, will be 3.5 mt, as I said, from the present around 0.5 mt.

In fact, at Surda, we will expand an existing running mine while Kendadih and Rakha will be re-opened mines. We had to close these 2 because of financial unviability. You see, LME prices had fallen beyond a limit and we had not been able to sustain the prices due to our higher production cost. In fact, all the mines in Ghatshila had to

be closed from 1997 till 2000 – Musabani, Rakha, Kendadih and Surda. The last was put on care and maintenance so that we could increase production when things started looking up.

The Ghatshila plant is more than 70 years old. Here, too the 20,000 tons capacity of the refinery will be expanded to 25,000 tons per annum. Similarly, the smelter's rated capacity of 18,500 tons per annum will be enhanced to match with the refining capacity, to 25,000 tons per annum.

We need to start mine equipping afresh. We have already started the job of reopening Kendadih within the next 6 months. Mine equipping has begun and the bids for mining contracts for an MDO are in position. The Rakha mine will also be re-opened after mine equipping through an MDO.

But Chhapri-Siddheswari are 2 new mines and thus these will take a little more time to start. Here, we plan to take up development of Rakha as well as Chhapri-Siddheswari through a mine-developer-operator (MDO). We will enter into an arrangement with the MDO whereby we will buy the metal-in-concentrate (MIC) from the former. We plan to appoint the MDO within this fiscal. We have completed one round of bidding where 8 parties had participated. They have suggested certain modifications in the tender which we are incorporating. These will be vetted by our finance and other relevant departments and then the final round will take place soon.

Both overseas and domestic players participated in the bids.

Another 5 mt will come from Malanjkhand. Here, as I have already mentioned, the mining expansion will be from the present 2 mt to 5 mt per annum. The beneficiation plant of 2 mt capacity will also be expanded – another 3 mt capacity will be added.

These additions will take up our mining capacity to 12 mt in from the present 3 mt. However, this cannot be achieved at one go but, gradually, in stages.

opencast mine will be closed in a phased manner and we will start underground mining of 5 mt in the same mine. It is like a bowl with the open cast on top and the process of developing the underground mine right under the opencast is already on – it is expected that by 2020 the project will be completed and commercial production will start from 2018 itself. Once production from the opencast starts tapering off, we will start ramping up output from the underground mine.

You said HCL will also be expanding its beneficiation capacity...

Yes, we will expand the Malanjkhand beneficiation capacity from the present 2 mt capacity to 5 mt and at Rakha we will construct a beneficiation plant of 3 mt capacity because the existing beneficiation capacity is of only 0.9 mt whereas we are required to process ore to the tune of 3.5 mt.

We already have a beneficiation plant of 1.5 mt at Khetri, which we will hike to 3.5 mt. We will expand that too but after these expansions are in place. So, our present beneficiation capacity is 4.5 mt and it has to come up to 12 mt to match up to our expanded mining capacity by 2022.

What are the smelter expansion plans like?

Our existing smelter at Ghatshila has a capacity of 18,500 tons per annum, as mentioned above. We have another very old smelter at Khetri which is not operational. We closed it because it was economically unviable. Therefore, we plan to set up a processing plant, I will not call it a smelter, which will process the same concentrate that is processed through the smelter but through a state-of-the-art hydro-metallurgical process. At present, the project is under feasibility stage.

The Jhagadia plant, which was acquired in 2015, has a capacity of 50,000 tons of cathode production per annum. At present,

25,000 tons is operational and the remaining 25,000 tons are under trial but by December 2017 the plant is likely to run to full capacity.

We have set up the Kaldo furnace at this plant which is a technology acquired from Boliden of Sweden. Only 11 such furnaces are operational worldwide and it is the only one in India. Kaldo has the ability to treat all the copper secondary and waste materials having low sulphur content, along with electronic wastes. In short, whatever is not treated by a conventional smelter is treated by this furnace in an environment-friendly manner to produce refined copper of LME grade.

Importantly, financially also, the Kaldo furnace will create value. The impurities in a normal smelter costs us money but through Kaldo we can earn revenues by processing the impurities through a beneficiation process.

So, do you plant to build a waste heat recovery plant?

The waste heat recovery plant is inbuilt into the design at Jhagadia. It is a zero discharge plant. We preheat and vapourise the content, the solid is collected and refined. We have a fully environment-compliant effluent treatment plant at Jhagadia.

In fact, Kaldo has the distinction of being able to process any material – copper, silver, gold, cadmium, etc.

Is your Jhagadia plant the only of its kind in India?

Yes, it can use imported copper concentrate as well as low-grade copper material. Our existing smelters cannot process any material with less than 20-25 percent copper. The Kaldo can process even less than 12 percent copper.

Importantly, there are many wastes generated at intermediary stages of copper production at our plants in the form of slag, reverts, copper jam etc. which can also be processed by Kaldo.

Are you looking to close down any unviable mines, just as you are re-opening a few?

No, none at present. However, let me add that the Malanjkhand 2 mt capacity

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Where mining is concerned, HCL's current ore production capacity is 3 mt per annum but we have plans to increase this to 12 mt by 2022.

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Traditionally, we get 15-20 percent of secondaries. If we process these through the conventional method, it will cost us much more. But Kaldo can process all low-copper-bearing material in a profitable manner.

How much of intermediaries is HCL generating?

We generate around 3,000 tons of intermediaries per annum. Where the remaining material is concerned, we can buy or use someone else's material. We are working on all three models. We use our intermediaries, buy imported copper concentrate and go in for tolling whereby a third party with impure material can get it processed at our plant to produce copper cathodes.

We take a refining charge in the third model. So, there is a flexibility in our business operations.

Additionally, there is an advantage with the Jhagadia plant in that the copper cathodes produced here are LME price neutral because of the imported concentrate used. Even in case of low LME prices scenario, the plant will be highly viable for the company.

How much of imported copper concentrate does HCL use?

We will utilise less than 50 percent of the Kaldo furnace's capacity through imports because we have our own intermediaries (3,000 tons produced from our plants) and we will do 25,000 tons through the tolling route.

Are you looking at any new business areas?

Yes, we are looking at waste-to-wealth projects as the next focus areas. We produce mining waste in the form of tailings.

These occur after we take out the copper concentrate. Only 1 percent copper is present in the copper ore, which is called metal-in-concentrate. We convert this concentrate into powder which contains 25 percent copper. The remaining 99 percent of the ore is waste material or tailings, which contain not only some amount of copper but precious metals like gold and silver too! Till date we did not have any technology to extract these metals from the copper tailings

We have set up the Kaldo furnace at this plant which is a technology acquired from Boliden of Sweden. Only 11 such furnaces are operational worldwide and it is the only one in India. Kaldo has the ability to treat all the copper secondary and waste materials having low sulphur content, along with electronic wastes. In short, whatever is not treated by a conventional smelter is treated by this furnace in an environment-friendly manner to produce refined copper of LME grade.

But, we are now working on a project called Copper Ore Tailing, at Malanjkhanda where we will treat 3.29 mt per annum or 10,000 tons of tailings per day. The plant's construction is expected to be over by December and commissioning is likely by March of this fiscal. This project is likely to fetch us revenues of around ₹150 crore per annum.

Importantly, through this project, we will be able to separate and produce gold, silver, copper and glass-grade silica.

And we will replicate the same plant at Khetri and Ghatshila at a later stage, after judging the success of this one. Actually, a 200 tons per day tailings treatment pilot plant at Khetri has proved that these precious metals can be extracted. Based on its success story we decided to build the new plant at Malanjkhanda.

We have more than 100 mt stocks of tailings – 50 mt at Malanjkhanda and 50 mt at Khetri. So, the advantage is we already have the raw materials and now only have to prove our efficiency in processing in this parallel line of business, where Hindustan Copper can actually become a producer of precious metals as well.

Are you looking at any metals/minerals other than gold, silver and silica?

Yes, in fact, we are looking long haul at the new business – even looking to recover these metals from the refineries as well. We have started recovering nickel. There are several new business areas.

One, we are looking at creating value out of the waste rock we produce, an area that is yet to be explored. You see, we generate a lot of waste rock during

mining. Say, we mine 100 kg, the copper ore will be 25 kg and the balance will be waste. And that 25 percent will contain 1 percent copper, and here too the rest will be waste. Hence, the stress on beneficiation and refining to extract as much value from the waste as possible, as I have already discussed above. In fact, we have an inventory of 100 mt of this waste rock which is very hard material and can be used as railway ballast aggregate, in the construction industry as well as crushed to form sand. However, since it is not an off-the-shelf product, marketing it will be an issue. So, we may forge a joint venture with a private player, regarding this business.

Second is nickel. Our refinery refines to give us copper to 99.9 percent purity. In the process we produce some wastes like "bleed" from the refinery plant. To explain further, bleeding is done to take out the impurities. The solution inside the refinery receives impurities but after some time the level of impurities goes beyond the treatment limit and we have to bleed it to bring in fresh solution. This refinery bleed is actually sulphuric acid, which is damaging to the environment. Before taking it out, we neutralise it with caustic soda but this is a cost to us. Moreover, materials like nickel and copper also go out with this bleed. So, we are losing nickel and copper to a certain extent, which is an added cost!

So, by installing a nickel plant at Ghatshila, we would be able to extract nickel present in the waste solution and copper and also the sulphuric acid for re-circulation.

Third is glass-grade silica. The Malanjkhanda mine has a lot of this deposit. If we treat 10,000 tons of copper tailings per

day, 50 percent of this will be glass-grade silica, which can also be used in construction and the glass factories. We have not factored in revenues from silica and waste rock and are discussing with players like MMTC on selling the same.

Fourth, the slag we generate during the process of manufacturing copper also has copper and iron ore traces. We are doing R&D to get value out of this too.

There is scope for e-auction of these materials...

Yes, there is huge scope for selling these through this route.

So, how much can all your expansion plans and new business areas and tie-ups contribute to HCL's additional revenues?

The new businesses are likely to generate additional revenues of ₹55-60 crore per annum.

By 2020, our turnover, which was at ₹1,216 crore in the last fiscal, should be touching ₹5,000 crore.

We also aim to invest ₹3,000 crore in our expansions till 2020 and this will be funded through a long-term term loan and equity. We plan to raise ₹1,000 crore through equity and the balance amount through debt.

So, 2020 will be a benchmark year for HCL...

Yes, in the coming 2 years, all our synergies will be focused on the new business areas. By the end of this fiscal we are targeting a turnover of ₹2,000 crore.

Are you looking any newer mines, going forward?

We have already applied for area reservation for exploration of copper mineral. For instance, we are entering into a JV with the Chhattisgarh Mineral Development Corporation for exploration of copper in that area. The mining potential is yet to be identified... we need to undertake exploration... The area is 60 km from Raipur and it should be a 2-2.5 mt mine. So let us see... We are in the process of forming the JV. We expect the ore quality to match that in Malankhand, of 1 percent.

Here too, we will take the MDO route. We can also take MECL's help in exploring the mine here.

Are you looking at other similar tie-ups and JVs?

At present, no. But, going forward, if there are opportunities in the waste-to-wealth segment, then we could consider. For instance, in waste rock, under our new business activities.

What is the scenario like with regard to the copper industry?

The copper industry is likely to grow by about 8 percent as per estimates this year. The country produces 8 lakh tons of the metal. And we consume around 6.5 lakh tons. HCL's share in this is 5 percent which is likely to increase to 15 percent. But HCL is an integrated player. We are capturing almost 90 percent of the value on LME and doing only 10 percent value-addition. Where we value-add, the volume is less but we capture maximum value. Our prices are benchmarked to the LME, which is not the case with iron ore, which gets a protected zone to operate within.

Moreover, if 100 kg of iron ore is mined, 80 kg will be extractable ore unlike in copper, where only 1 percent is extractable ore. Moreover, it is the hardest form of ore. Processing of harder rock means more production cost. HCL's prices are linked to the LME and so we have to weather the fluctuations as well.

Is there any drawback to the fact that HCL's prices are linked to the LME other than the volatility?

The country produces 8 lakh tons and consumes 6.5 lakh tons. The country exports 2.5 lakh tons but imports around 3 lakh tons. There is cross-trading here, which is hurting us. Now imports have captured 30 percent of the market. By virtue of the free trade agreements, imports are coming into India from ASEAN countries at zero duty, and at 1 percent from Japan.

Copper is exported to China from India. The imports were a bit costly earlier but have now become cheaper because of the FTAs with ASEAN and Japan. By virtue

of that margin of say 2 percent, which was there previously on a rate of ₹50 per ton, this could be hiked to ₹51 per ton. That margin is now lost.

The market is competitive and, at times, we have to sell at a discount to LME prices. We cannot get the full premium that the market used to demand. So, at times, we have to sell at the lower end of the LME.

What factors could propel domestic copper demand growth?

A lot of good initiatives have been taken by the central government – Make in India, Housing for All, Metro Rail and Smart Cities, Dedicated Freight Corridors – these are bound to boost the demand for copper. The green shoots are already visible.

What is the outlook on copper prices?

Currently, LME prices are at around \$6,500 per ton. Where the outlook is concerned, we expect the prices to go down to \$6,000 per ton by this year-end. I don't expect these to go down further than this. Prices were hovering around \$6,700 per ton earlier. Demand and dollar fluctuations are reasons for the slide.

The role of HCL in terms of prices in India is that of a market regulator. The small buyers buy from us. The private copper producers would have otherwise hiked prices to maximise their profit. But, because of HCL's pricing, they are unable to do that. Volume-wise, we have only 5 percent market share at present. But, price-wise, we do play a definite role.

What is your product portfolio like at present?

We produce metal in concentrate, cathodes and rods. The maximum contribution to revenues comes from MIC, then cathodes and rods. Around 90 percent of revenue contribution comes from refined copper but we have a lot flexibility in our operations because we produce 32,000 tons of the refined metal and process around 20,000 tons per annum. This production will improve to 50,000 tons when the Jhagadia plant becomes fully operational this year.

By 2020, our metal production will reach 100,000 tons.

# Prices of Steel products in various markets

Steel Insights Bureau

5V SHU WRQ

| Product                | Name of the Market/<br>SRGEK•ECVKQP | Price Rs./Ton<br>as on<br>3-Nov-17 | Price Rs./Ton<br>as on<br>2-Nov-17 | Price Rs./Ton<br>as on<br>26-Oct-17 | Price Rs./Ton<br>as on<br>4-Oct-17 | Price Rs./Ton<br>as on<br>6-May-17 | Monthly<br>changes |
|------------------------|-------------------------------------|------------------------------------|------------------------------------|-------------------------------------|------------------------------------|------------------------------------|--------------------|
| ,1*27                  | 0\$1', *29,1'*\$5+                  |                                    |                                    |                                     |                                    |                                    | X                  |
| ,1*27                  | \$/ \$1*                            |                                    |                                    |                                     |                                    |                                    | X                  |
| ,1*27                  | 080%\$,                             |                                    |                                    |                                     |                                    |                                    | X                  |
| 6&5\$3                 | 0\$1', *2%,1'*\$5+                  |                                    |                                    |                                     |                                    |                                    | X                  |
| 6&5\$3                 | \$/ \$1* 6&5\$3 <\$5'               |                                    |                                    |                                     |                                    |                                    | X                  |
| 6&5\$3                 | 080%\$,                             |                                    |                                    |                                     |                                    |                                    | X                  |
| 0(/,1* 6&5\$3          | 0\$1', *2%,1'*\$5+                  |                                    |                                    |                                     |                                    |                                    | N/A                |
| \$1*/(                 | 0\$1', *2%,1'*\$5+                  |                                    |                                    |                                     |                                    |                                    | X                  |
| %,/(7                  | 0\$1', *2%,1'*\$5+                  |                                    |                                    |                                     |                                    |                                    | X                  |
| %,/(7                  | '85*\$385                           |                                    |                                    |                                     |                                    |                                    | X                  |
| 6321*(,521             | '85*\$385                           |                                    |                                    |                                     |                                    |                                    | X                  |
| 6321*(,521             | 5\$,*\$5+                           |                                    |                                    |                                     |                                    |                                    | X                  |
| 3,* ,521               | 7\$/\$                              |                                    |                                    |                                     |                                    |                                    | Z                  |
| 3,* ,521               | -\$/\$1'+\$5                        |                                    |                                    |                                     |                                    |                                    | Z                  |
| 707 PP                 | &+(11\$,                            |                                    |                                    |                                     |                                    |                                    | X                  |
| 707 PP                 | %+\$5\$70\$7\$                      |                                    |                                    |                                     |                                    |                                    | Z                  |
| 707 PP                 | *<\$1                               |                                    |                                    |                                     |                                    |                                    | Z                  |
| 707 PP                 | -<27,                               |                                    |                                    |                                     |                                    |                                    | X                  |
| +5 &2,/6               | '(/+,                               |                                    |                                    |                                     |                                    |                                    | Ž                  |
| +5 &2,/6               | .2/.\$7\$                           |                                    |                                    |                                     |                                    |                                    | Ž                  |
| &5 &2,/6               | 080%\$,                             |                                    |                                    |                                     |                                    |                                    | X                  |
| &5 &2,/6               | /8'+,\$1\$                          |                                    |                                    |                                     |                                    |                                    | Ž                  |
| ,521 25( 10'&          | PP                                  |                                    |                                    |                                     |                                    |                                    | Z                  |
| ,521 25( 10'&          | PP                                  |                                    |                                    |                                     |                                    |                                    | Z                  |
| ,521 25( 10'&          | ),1(6                               |                                    |                                    |                                     |                                    |                                    | Z                  |
| ,521 25( %(/\$5<       | PP PP                               |                                    |                                    |                                     |                                    |                                    | Z                  |
| ,521 25( 7\$7\$1\$*\$5 | PP PP                               |                                    |                                    |                                     |                                    |                                    | Ž                  |
| ,521 25( %\$5%,/       | PP                                  |                                    |                                    |                                     |                                    |                                    | Z                  |
| &+\$11(/               | *+\$=,\$%\$'                        |                                    |                                    |                                     |                                    |                                    | X                  |
| &+\$11(/               | 5\$,385                             |                                    |                                    |                                     |                                    |                                    | X                  |
| .,5( 52'               | '85*\$385                           |                                    |                                    |                                     |                                    |                                    | X                  |
| .,5( 52'               | 5\$,385                             |                                    |                                    |                                     |                                    |                                    | X                  |

1 R W Prices are basic, exclusive of taxes

Iron ore prices are FOR, excluding Royalty, DMF, NMET, Cess, Forest Permit Fee etc.

HR Coil &amp; CR Coil prices are exclusive of ED &amp; taxes

# Ferro alloys & metals price trends

Steel Insights Bureau

| Ferro Alloys           | October 2017 | September 2017 | August 2017 |
|------------------------|--------------|----------------|-------------|
| ([ ZRUNV 5V WRQ        |              |                |             |
| )HUUR 6LOLFRQ 6L       |              |                |             |
| +& )HUUR &KURPH &U     |              |                |             |
| +& )HUUR 0DQJDQHVH 0Q  |              |                |             |
| 6LOLFR 0DQJDQHVH 0Q 6L |              |                |             |
| )HUUR 7LWDQLXP 7L      |              |                | 1 \$        |
| ([ ZRUNV 5V NJ         |              |                |             |
| )HUUR 9DQDGLXP 9       |              | 1 \$           | 1 \$        |
| )HUUR 0RO\ 0R PLQ 1 \$ |              | 1 \$           | 1 \$        |

# Base metals price trends

Steel Insights Bureau

| Non Ferrous Metals        | October 2017 | September 2017 | August 2017 |
|---------------------------|--------------|----------------|-------------|
| ([ ZRUNV 5V WRQ           |              |                |             |
| Aluminium                 |              |                |             |
| ([ ZRUNV ([ *RGRZQ 5V WRQ |              |                |             |
| &RSSHU                    |              |                |             |
| 86' 7RQ                   |              |                |             |
| /HDG                      |              |                |             |
| Zinc                      |              |                |             |
| 7LQ                       |              |                |             |
| 1LFNHO                    |              |                |             |

# Production, imports, exports, availability & apparent consumption (provisional) April 2017 - September

Steel Insights Bureau

(in '000 tons)

| PRODUCERS                   | FINISHED STEEL           |                      |             |                      |                      |             |                      |                      |             |
|-----------------------------|--------------------------|----------------------|-------------|----------------------|----------------------|-------------|----------------------|----------------------|-------------|
|                             | Non-Alloy Steel (Carbon) |                      |             | Alloy Steel          |                      |             | Total                |                      |             |
|                             | 2017 - 18<br>(Prov.)     | 2016 - 17<br>(Prov.) | % Variation | 2017 - 18<br>(Prov.) | 2016 - 17<br>(Prov.) | % Variation | 2017 - 18<br>(Prov.) | 2016 - 17<br>(Prov.) | % Variation |
| 6 \$ , /                    |                          |                      |             |                      |                      |             |                      |                      |             |
| 5 , 1 /                     |                          |                      |             |                      |                      |             |                      |                      |             |
| 7 6 /                       |                          |                      |             |                      |                      |             |                      |                      |             |
| ESSAR                       |                          |                      |             |                      |                      |             |                      |                      |             |
| - 6 : /                     |                          |                      |             |                      |                      |             |                      |                      |             |
| - 6 3 /                     |                          |                      |             |                      |                      |             |                      |                      |             |
| 2 7 + ( 5 6                 |                          |                      |             |                      |                      |             |                      |                      |             |
| *URVV 3URGFWLRQ             |                          |                      |             |                      |                      |             |                      |                      |             |
| /HVV ,37 2ZQ &RQVXPSWLRQ    |                          |                      |             |                      |                      |             |                      |                      |             |
| F 7RWDO 3URGFWLRQ IRU 6DOH  |                          |                      |             |                      |                      |             |                      |                      |             |
| G ,PSRUWV                   |                          |                      |             |                      |                      |             |                      |                      |             |
| H ([SRUWV                   |                          |                      |             |                      |                      |             |                      |                      |             |
| H \$YDLODELOLW\ F G H       |                          |                      |             |                      |                      |             |                      |                      |             |
| I 9DULDWLRQ LQ 6WRFN        |                          |                      |             |                      |                      |             |                      |                      |             |
| J \$SSDUHQW &RQVXPSWLRQ H I |                          |                      |             |                      |                      |             |                      |                      |             |
| /HVV 'RXEOH &RXQLQJ         |                          |                      |             |                      |                      |             |                      |                      |             |
| &RQVXPSWLRQ                 |                          |                      |             |                      |                      |             |                      |                      |             |

6 R X U Steel Ministry

# Sponge iron production for September 2017

Steel Insights Bureau

(In '000 tons)

| Producers                | SEPTEMBER            |                      |             | SEPTEMBER Vis - A - Vis AUGUST |                     |             | APRIL - SEPTEMBER    |                      |             |
|--------------------------|----------------------|----------------------|-------------|--------------------------------|---------------------|-------------|----------------------|----------------------|-------------|
|                          | 2017 - 18<br>(Prov.) | 2016 - 17<br>(Prov.) | % Variation | SEPTEMBER<br>2017 (Prov.)      | AUG 2017<br>(Prov.) | % Variation | 2017 - 18<br>(Prov.) | 2016 - 17<br>(Prov.) | % Variation |
| A ESSAR                  |                      |                      |             |                                |                     |             |                      |                      |             |
| % - 6 : /                |                      |                      |             |                                |                     |             |                      |                      |             |
| & - 63 /                 |                      |                      |             |                                |                     |             |                      |                      |             |
| ' 27 + (56               |                      |                      |             |                                |                     |             |                      |                      |             |
| * 5266 352'8&7,21        |                      |                      |             |                                |                     |             |                      |                      |             |
| /HVV ,37 2ZQ &RQVXPSWLRQ |                      |                      |             |                                |                     |             |                      |                      |             |
| 7RWDO 3URGXFWRQ IRU 6DOH |                      |                      |             |                                |                     |             |                      |                      |             |

6 R X U Steel Ministry

# Pig iron production for September 2017

Steel Insights Bureau

(in '000 tons)

| Producers                   | SEPTEMBER            |                      |             | SEPTEMBER Vis - A - Vis AUGUST |                           |             | APRIL - SEPTEMBER    |                      |             |
|-----------------------------|----------------------|----------------------|-------------|--------------------------------|---------------------------|-------------|----------------------|----------------------|-------------|
|                             | 2017 - 18<br>(Prov.) | 2016 - 17<br>(Prov.) | % Variation | SEPTEMBER<br>2017 (Prov.)      | AUGUST<br>2017<br>(Prov.) | % Variation | 2017 - 18<br>(Prov.) | 2016 - 17<br>(Prov.) | % Variation |
| \$ 6 \$, /                  |                      |                      |             |                                |                           |             |                      |                      |             |
| D %KLODL 6WHHO 3ODQW        |                      |                      |             |                                |                           |             |                      |                      |             |
| E 'XUJDSXU 6WHHO 3ODQW      |                      |                      |             |                                |                           |             |                      |                      |             |
| F 5RXUNHOD 6WHHO 3ODQW      |                      |                      |             |                                |                           |             |                      |                      |             |
| G %RNDUR 6WHO 3ODQW         |                      |                      |             |                                |                           |             |                      |                      |             |
| H ,QGLDQ ,URQ 6WHHO 3ODQW   |                      |                      |             |                                |                           |             |                      |                      |             |
| I \$OOR\ 6WHHO 3ODQW        |                      |                      |             |                                |                           |             |                      |                      |             |
| J 663                       |                      |                      |             |                                |                           |             |                      |                      |             |
| K 9LV ,URQ 6WHHO            |                      |                      |             |                                |                           |             |                      |                      |             |
| 7RWDO 6 \$, /               |                      |                      |             |                                |                           |             |                      |                      |             |
| % 5,1/ 963                  |                      |                      |             |                                |                           |             |                      |                      |             |
| & 76 /                      |                      |                      |             |                                |                           |             |                      |                      |             |
| ' (66\$5                    |                      |                      |             |                                |                           |             |                      |                      |             |
| ( - 6 : /                   |                      |                      |             |                                |                           |             |                      |                      |             |
| ) - 63 /                    |                      |                      |             |                                |                           |             |                      |                      |             |
| * 27 + (56                  |                      |                      |             |                                |                           |             |                      |                      |             |
| * 5266 352'8&7,21           |                      |                      |             |                                |                           |             |                      |                      |             |
| /HVV ,37 2ZQ<br>&RQVXPSWLRQ |                      |                      |             |                                |                           |             |                      |                      |             |
| 7RWDO 3URGXFWRQ IRU 6DOH    |                      |                      |             |                                |                           |             |                      |                      |             |

6 R X U Steel Ministry

# Import, export, availability, stock & consumption iron & steel Apr-Sept 2017 (provisional)

Steel Insights Bureau

4 X D Q W L W \ μ WR Q

| CATEAGORY                    | Total<br>Production<br>For Sale | Imports | Exports | Availability | Stock                    |                          |                       | Apparent<br>Consumption      |                             | Double Count                 |                             | Consumption                  |                             | Consumption<br>Variation<br>Over<br>Last Year(%) |
|------------------------------|---------------------------------|---------|---------|--------------|--------------------------|--------------------------|-----------------------|------------------------------|-----------------------------|------------------------------|-----------------------------|------------------------------|-----------------------------|--------------------------------------------------|
|                              |                                 |         |         |              | As on<br>01-APR-<br>2017 | As on<br>30-SEP-<br>2017 | Variation<br>in Stock | Current<br>Year<br>(2017-18) | Last Year<br>(2016 -<br>17) | Current<br>Year<br>(2017-18) | Last Year<br>(2016 -<br>17) | Current<br>Year<br>(2017-18) | Last Year<br>(2016 -<br>17) |                                                  |
| \$ 3 , * , 5 2 1             |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| % 6 3 2 1 * ( , 5 2 1 ' 5 ,  |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| & 6 ( 0 , 6 I R U 6 D O H    |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| FINISHED STEEL (NON - ALLOY) |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| % D U V 5 R G V              |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| 6 W U X F W X U D O V        |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| 5 O \ Materials              |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| 7 R W D O 1 R Q ) O D W W R  |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| 3 O D W H V                  |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| + 5 & R L O V ? 6 N H O S    |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| + 5 6 K H H W V              |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| & 5 6 K H H W V ? & R L O V  |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| * 3 ? * & Sheets             |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| ( O H F 6 K H H W V          |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| 7 L Q S O D W H L Q F O Z Z  |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| 7 0 % 3                      |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| 3 L S H V / D U J H ' L D    |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| 7 L Q I U H H steel          |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| 7 R W D O ) O D W W R        |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| 7 2 7 \$ / 1 R Q \$ O O R \  |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| FINISHED STEEL (ALLOY)       |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| 1 R Q ) O D W                |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| ) O D W                      |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| 7 R W D O \$ O O R \         |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |
| * 5 \$ 1 ' 7 2 7 \$ /        |                                 |         |         |              |                          |                          |                       |                              |                             |                              |                             |                              |                             |                                                  |

6 R X U Steel Ministry

# Imports of iron & steel through major Indian ports in Apr-Sept 2017 vis-a-vis Apr-Sept 2016

Steel Insights Bureau

4 X D Q W L W \ μ W R Q

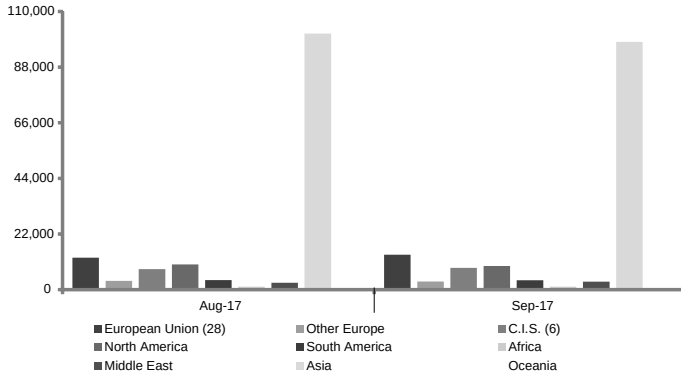
| Sl. No. | Category                                                         | Non-Alloy (Prime + Defective) | Non-Alloy (Prime + Defective) | Growth |
|---------|------------------------------------------------------------------|-------------------------------|-------------------------------|--------|
|         |                                                                  | Apr-Sep 17                    | Apr-Sep 16                    | %      |
| 1       | Steel                                                            |                               |                               |        |
|         | \$ 6 (0,6                                                        |                               |                               |        |
|         | % L O O H W V 6 O D E V H W F                                    |                               |                               |        |
|         | 5 H U R O O D E O H 6 F U D S                                    |                               |                               |        |
|         | % ) L Q L V K H G 6 W H H O                                      |                               |                               |        |
|         | 1 R Q ) O D W 3 U R G X F W V                                    |                               |                               |        |
|         | % \$ 5 6 5 2 ' 6                                                 |                               |                               |        |
|         | 6 7 5 8 & 7 8 5 \$ / 6                                           |                               |                               |        |
|         | 5 / < 0 \$ 7 ( 5 , \$ / 6                                        |                               |                               |        |
|         | 7 2 7 \$ / 1 R Q ) O D W 3 U R G X F W V                         |                               |                               |        |
|         | ) O D W 3 U R G X F W V                                          |                               |                               |        |
|         | 3 / \$ 7 ( 6                                                     |                               |                               |        |
|         | + 5 6 + ( ( 7 6                                                  |                               |                               |        |
|         | + 5 & 2 , / 6 7 5 , 3                                            |                               |                               |        |
|         | & 5 & 2 , / 6 + ( ( 7 6                                          |                               |                               |        |
|         | * 3 * & 6 + ( ( 7 6 & 2 , /                                      |                               |                               |        |
|         | ( / ( & 7 6 + ( ( 7 6                                            |                               |                               |        |
|         | 7 0 % 3                                                          |                               |                               |        |
|         | 7 , 1 3 / \$ 7 ( 6                                               |                               |                               |        |
|         | 7 , 1 ) 5 ( ( 6 7 ( ( /                                          |                               |                               |        |
|         | 3 , 3 ( 6                                                        |                               |                               |        |
|         | 7 2 7 \$ / ) O D W 3 U R G X F W V                               |                               |                               |        |
|         | 7 2 7 \$ / ) L Q L V K H G 6 W H H O 1 2 1 \$ / / 2 <            |                               |                               |        |
|         | 7 2 7 \$ / 6 W H H O \$ % 1 2 1 \$ / / 2 <                       |                               |                               |        |
|         | \$ O O R \ 6 W D L Q O H V V                                     |                               |                               |        |
|         | ) L Q L V K H G 6 W H H O                                        |                               |                               |        |
|         | Semis                                                            |                               |                               |        |
|         | 7 2 7 \$ / 6 7 ( ( / \$ / / 2 <                                  |                               |                               |        |
|         | 7 2 7 \$ / ) , 1 , 6 + ( ' 6 7 ( ( / \$ / / 2 < 1 2 1 \$ / / 2 < |                               |                               |        |
|         | 7 2 7 \$ / 6 7 ( ( / \$ / / 2 < 1 2 1 \$ / / 2 <                 |                               |                               |        |
| 2       | 2 W K H U 6 W H H O , W H P V                                    |                               |                               |        |
|         | ) , 7 7 , 1 * 6                                                  |                               |                               |        |
|         | 0 , 6 & 6 7 ( ( / L 7 ( 0 6                                      |                               |                               |        |
|         | 6 & 5 \$ 3                                                       |                               |                               |        |
| 3       | , U R Q                                                          |                               |                               |        |
|         | 3 , * , 5 2 1                                                    |                               |                               |        |
|         | 6 3 2 1 * ( , 5 2 1                                              |                               |                               |        |
|         | + % , 5 2 1                                                      |                               |                               |        |
| 9       | ) H U U R \$ O O R \ V                                           |                               |                               |        |
|         | * 5 \$ 1 ' 7 2 7 \$ /                                            |                               |                               |        |

6 R X U Steel Ministry

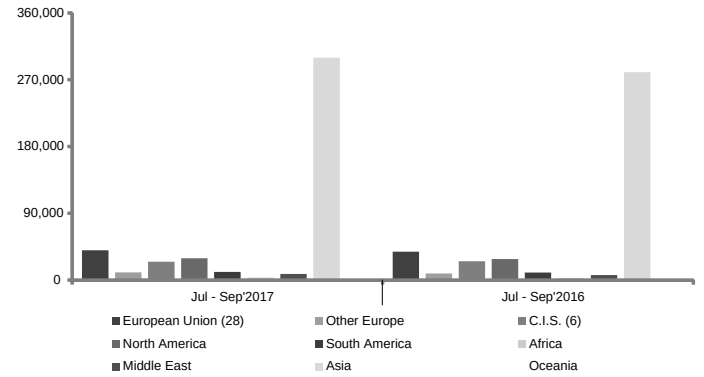
# World & Indian steel production data

Steel Insights Bureau

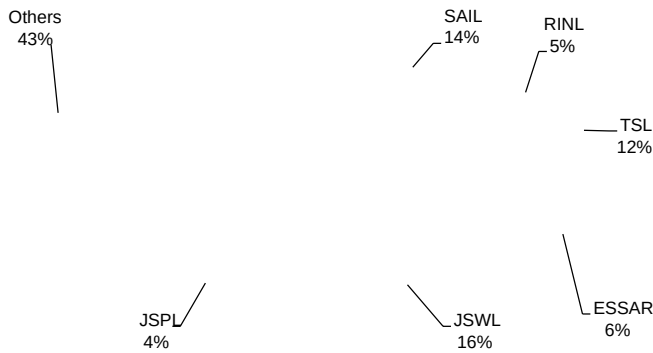
World crude steel production per region  
August 2017 vs September 2017



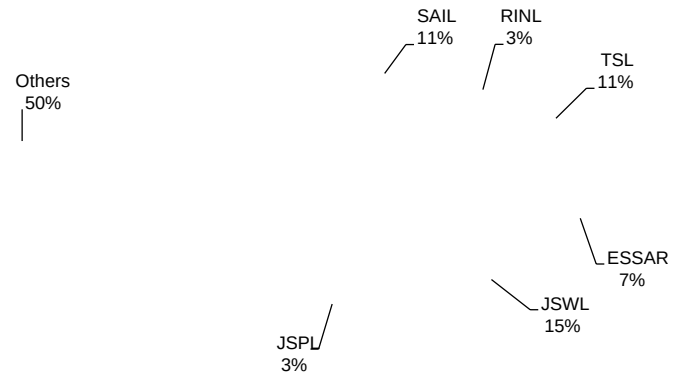
World crude steel production per region  
cumulative 2017 vs 2016



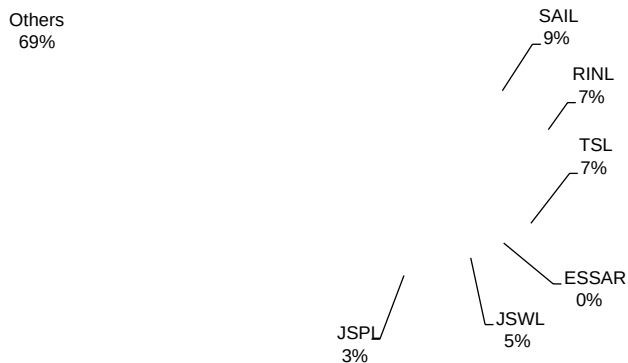
Crude Steel Production



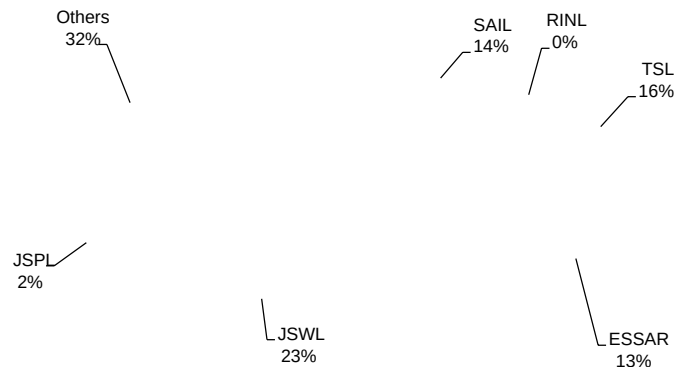
Finished Steel for Sale (Non - Alloy & Alloy)



Non - Flat Production for Sale (Non-Alloy & Alloy)



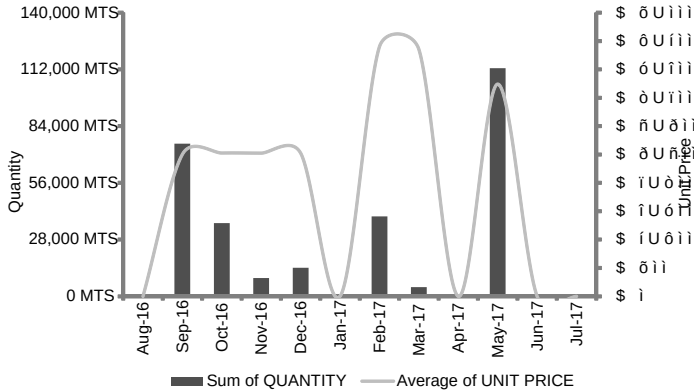
Flat Production for Sale (Non-Alloy & Alloy)



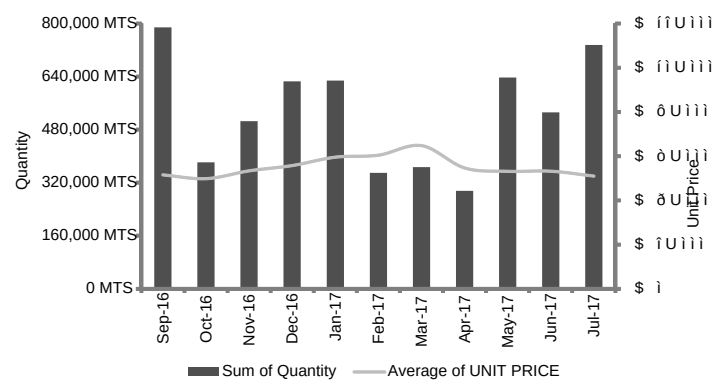
# Iron ore & ferro alloy import-export trends

Steel Insights Bureau

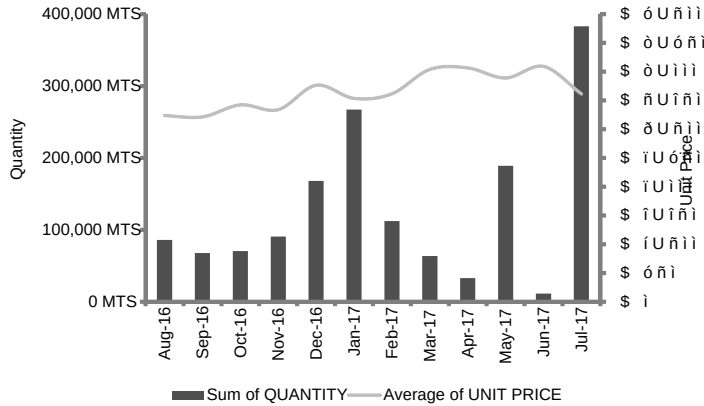
Iron ore pellets import price trend



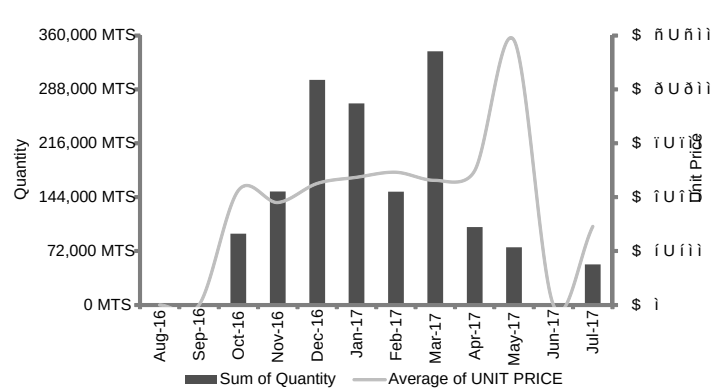
Iron ore pellets export price trend



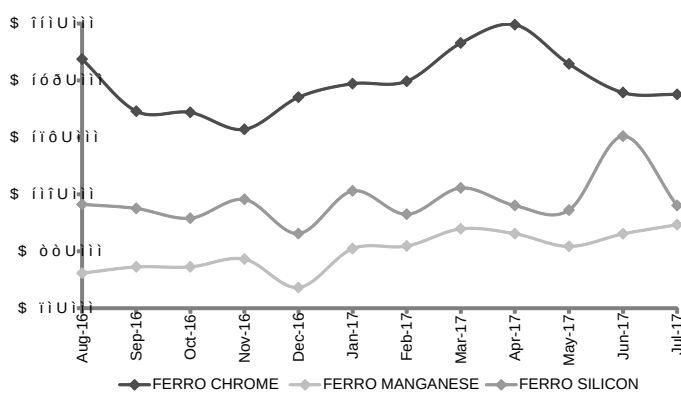
Iron ore lumps import price trend



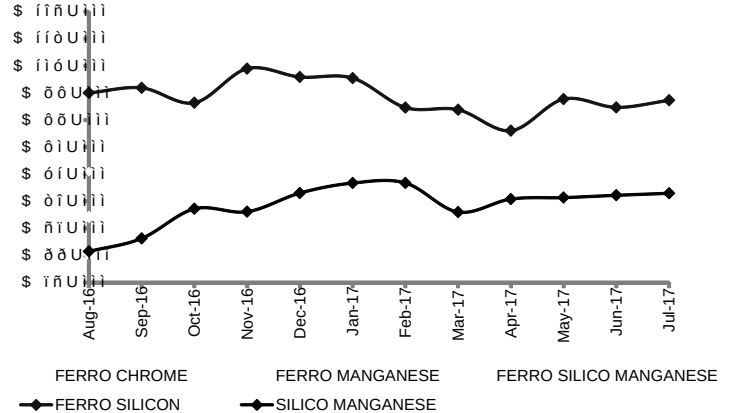
Iron ore lumps export price trend



Ferro alloy import price trend



Ferro alloy export price trend



\*For details break-up, subscribe India Steel Market Watch (ISMW)



# Export of Iron & Steel through major Indian ports April-September'17 vis-à-vis April-September'16

Steel Insights Bureau

4 X D Q W L W \ μ W R Q

| Sl. No. | Category                                                         | Non-Alloy (Prime + Defective) | Non-Alloy (Prime + Defective) | Growth |
|---------|------------------------------------------------------------------|-------------------------------|-------------------------------|--------|
|         |                                                                  | Apr-Sep 17                    | Apr-Sep 16                    | %      |
| ,       | Steel                                                            |                               |                               |        |
|         | \$ 6 ( 0 , 6                                                     |                               |                               |        |
|         | % L O O H W V 6 O D E V H W F                                    |                               |                               |        |
|         | 5 H U R O O D E O H 6 F U D S                                    |                               |                               |        |
|         | % ) L Q L V K H G 6 W H H O                                      |                               |                               |        |
|         | 1 R Q ) O D W 3 U R G X F W V                                    |                               |                               |        |
|         | % \$ 5 6 5 2 ' 6                                                 |                               |                               |        |
|         | 6 7 5 8 & 7 8 5 \$ / 6                                           |                               |                               |        |
|         | 5 / < 0 \$ 7 ( 5 , \$ / 6                                        |                               |                               |        |
|         | 7 2 7 \$ / 1 R Q ) O D W 3 U R G X F W V                         |                               |                               |        |
|         | ) O D W 3 U R G X F W V                                          |                               |                               |        |
|         | 3 / \$ 7 ( 6                                                     |                               |                               |        |
|         | + 5 6 + ( ( 7 6                                                  |                               |                               |        |
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